Dynamics, the Action Principle, and the Kinematics That Supports Them

From Quaternions to the Einstein–Yang–Mills Dumbbell, Verified in Lean 4

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Preface

This book is organised around one sentence: *a shape is kinematics; putting in an action produces dynamics.* Every chapter first shows a shape, then states the action, then derives the equations, and finally points to a theorem that has been checked by the Lean 4 proof assistant.

The shape that runs through the book is the *dumbbell*: two round caps joined by a cylindrical neck. With the Einstein–Yang–Mills action the caps become closed radiation universes and the neck becomes the throat of an extremal black hole, $\mathrm{AdS}_2 \times S^2$. Both have the same size ℓ, fixed by the gauge coupling through $\ell^2 = 4\pi G/g^2$. Before we get there we build everything we need from quaternions: rotations, spin, the three-sphere, Lorentz transformations, and the geometry of relativistic momentum.

The intended reader is an undergraduate in the second to fourth year who knows linear algebra, multivariable calculus and the first courses on mechanics and electromagnetism. General relativity, quantum field theory and group theory are not assumed; we build the parts we use.

How to read the status tags

Every result carries one of the following tags, taken directly from the claim ledger of the Lean repository.

proved in Lean	checked in Lean from its stated hypotheses
external theorem	uses a theorem from the literature, cited in the text
numerical	a numerical computation, not checked by Lean
negative result	an idea that was tried and does not work
open	not solved here (and in most cases not solved anywhere)

Physical readings ("this is a gravitational wave") are kept separate from theorems and are called interpretations. Exercises stay within what Lean has checked; each exercise names the theorem that answers it.

A guide for the specialist

This section is written for readers who already know general relativity and gauge theory and want to know, in two pages, what is in this book that they have not seen before, and how far it can be trusted.

What has been checked

The book is a companion to the Lean 4 repository *Cyclic Fueter Theory* (Zenodo concept DOI 10.5281/zenodo.23448911, version 0.6.0). At that version the repository contains 1873 theorem declarations in 171 modules. An axiom audit of 2421 entries (all public theorems and the index aliases) found no `sorry`; 2396 entries depend only on Lean's standard axioms `propext`, `Classical.choice` and `Quot.sound`, and the rest on a subset of them. Lean verifies each theorem from its declared hypotheses. It does not verify that the hypotheses describe nature, and results quoted from the literature are tagged external theorem.

Claims of novelty

The following statements are, to the author's knowledge, not stated in this form in the literature the author has consulted. The claim is deliberately weak: no systematic review has been carried out, no specialist has confirmed it, and several items are special cases or rearrangements of known results. They are listed so that they can be checked.

(1) **Linear rigidity** (Chapter 19). Every linear perturbation of the Einstein–Yang–Mills Tolman universe that preserves spherical symmetry, the magnetic ansatz and conformal flatness is generated by a conformal Killing vector of the Einstein static universe, so it is pure gauge.

(2) **Shape of the dumbbell** (Chapter 16). Time-symmetric, conformally flat slices satisfy the Bogomolny-type inequality $r'^2 \le W^2$. The case $W = \pm r'$ saturates it and gives $r^2 + \ell^2 r'^2 = \ell^2$. The cap is the regular solution of this equation and the Bertotti–Robinson neck is its singular solution, the envelope of the caps. The junction is fixed by continuity alone, is C^1 and never C^2, and the Hamiltonian constraint admits exactly the two values 0 and $-1/\ell$ of r'' there.

(3) **Null junction** (Chapters 16–17). A timelike junction between cap and neck is impossible, and the jump $[W_u] = 1/(\sqrt{2}\,\ell)$ is transported unchanged along the null fronts.

(4) **Weyl invariant** (Chapter 16). In this sector the Weyl invariant equals $3\big(2m/r - \ell^2(1 - W^2)^2/r^2\big)$, so conformal flatness says that the Misner–Sharp mass equals the magnetic self-energy on every sphere. This is a special case of the known relation between Weyl curvature and the Misner–Sharp mass.

(5) **One pair of constraints** (Chapter 23). The same Hamiltonian and momentum constraints reduce to the Friedmann constraint on the cap and to a flat two-dimensional Klein–Gordon equation on the Kantowski–Sachs side, where the neck $r = \ell$ is a fold of minisuperspace, invariant under $r \mapsto \ell^2/r$. The neck length is canonically conjugate to $-r\dot{r}$. Minisuperspace quantisation of Reissner–Nordström interiors is known; the reading in terms of the dumbbell is what is claimed.

(6) **Constant-area data** (Chapter 26). Time-symmetric data of constant areal radius r_0 obey $W'^2 = (1 - \ell^2(1 - W^2)^2/r_0^2)/(2\ell^2)$, with the lump $W = \sqrt{2}/\cosh(x/\ell)$ at $r_0 = \ell$. Only the neck keeps its area at the next instant.

(7) **Weak gravity conjecture** (Chapter 26). With an adjoint Higgs field the electric condition for the W boson and the magnetic condition for the BPS monopole coincide, $4\pi G v^2 \leq 1$. Together with the stability bound of the neck they give $3/4 \leq (m_W \ell)^2 \leq 1$, and the neck exists only for $\Lambda \leq g^2/(16\pi G)$.

(8) **Formalisation** (whole book). The spherically symmetric Einstein–Yang–Mills–Dirac sector is machine-checked from one reduced action through the constraints, the known limits and the thermodynamic relations.

What is not claimed

The cap and neck solutions themselves, the reduced action, the canonical form and the mass function, JT gravity near the neck, black-hole thermodynamics, the sphaleron zero mode, and the weak gravity and Festina Lente conjectures are standard. The general solution of the Wheeler–DeWitt equation with matter, the quantisation of gravitational waves beyond the linear level, and any derivation of the fine-structure constant are *not* obtained; Chapter 27 lists them as open problems.

Acknowledgements

The Lean proofs, numerical checks and parts of the text were prepared with the assistance of a large language model (Claude, Anthropic); the author has checked the results and takes responsibility for the content.

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Plan of the book

This preliminary draft contains the front matter and Chapter 16, which fixes the style of the whole book. The other chapters are planned as follows; the right-hand column names the main Lean modules each chapter draws on.

	Title	Main Lean modules	Status
Part I. Quaternions and rotations			
1	From complex numbers to quaternions	`AlphaI`	planned
2	Rotations and the three-sphere	`AlphaI`, `AlphaCompleteB`	planned
3	Fueter operators	`AlphaI_Operators`, `AlphaI_C`, `AlphaI_CFamily`	planned
4	Lorentz transformations: rotation $\times$ boost	`AlphaPillar_Polar`, `AlphaRoot_Duality`	planned
Part II. The quantum sphere and the mass shell			
5	The mass-shell hyperboloid	`AlphaB_QH`, `AlphaI_H3`	planned
6	The quantum sphere	`AlphaI_QBij`	planned
7	The Dirac equation and its projector	`AlphaV`, `AlphaV_FWOrder`	planned
8	Connections and curvature: Fueter, Berry, Wigner	`AlphaI_AW`, `AlphaV_Wigner`	planned
9	Monopoles and zero modes	`AlphaCompleteC`, `AlphaCompleteD`, `ZeroModeL2`	planned
Interlude A. The Standard Model from an internal sphere			
A	$SO(10)$, Pati–Salam, $B-L$ and chirality	`AlphaVII_*`, `AlphaFix_Gauge`, `AlphaFix_Z28`	planned
Part III. Curved space and field equations			
10	Moving frames and curvature	`Frame`, `AlphaS4_Curv`	planned
11	Slices and constraints	`AlphaII`, `AlphaIII_Codazzi`	planned
12	Gauge fields	`GaugeCovariance`, `AlphaIX_YM`, `AlphaVII_Maxwell`	planned
13	The action principle	`AlphaS1_Derivation`, `AlphaS2_*`, `AlphaIX_Classical`	planned
Part IV. The dumbbell			
14	Three phases	`AlphaII_*`, `AlphaFix_Geo`, `C2`	planned
15	Cap and neck	`AlphaII_EYM`, `AlphaIII_Glue`	planned
16	The shape of the dumbbell	`AlphaDumbbell_Shape`, `AlphaDumbbell_Bogomolny`, `AlphaJunction_Unique`	**drafted**
17	Junctions	`AlphaIII*`, `AlphaS4_*`, `AlphaReal_Junction`	planned
18	Time evolution	`AlphaIV*`, `AlphaS5_*`, `AlphaReal_Rest`	planned
19	The conformal frame and Ω	`AlphaPillar_Conformal`, `AlphaOmega_*`	planned
20	Stability	`AlphaXI_*`, `AlphaPillar_Smooth`	planned
Interlude B. Contact with string theory			

	Title	Main Lean modules	Status
B	WZW, the standard embedding and attractors	`AlphaVIII_*`	planned
Part V. Heat and quanta			
21	Black-hole thermodynamics	`AlphaIX_Thermo`, `AlphaSphericalClose`	planned
22	Quantum mechanics of the neck	`AlphaIX_JT`, `AlphaFamous`	planned
23	Quantum cosmology	`AlphaCanonical`, `AlphaQuantumSpherical`, `AlphaKS_WDW`	planned
24	Quantum fields in curved spacetime	`AlphaV_FRW`, `AlphaV_Anomaly`	planned
25	Fermions	`AlphaFermion`, `AlphaV_S3Dirac`	planned
Part VI. Shape, swampland and open problems			
26	Shape degrees of freedom and the weak gravity conjecture	`AlphaGeneral`, `AlphaConstArea`, `AlphaWGC`	planned
27	Open problems	`AlphaIdeas_Check`, ledger	planned

Part I

Quaternions and rotations

Chapter 1

From complex numbers to quaternions

The question

Complex numbers rotate the plane. Is there a number system that rotates space in the same way? There is, but it needs four components instead of three, and its multiplication is not commutative. That price turns out to be exactly what rotations demand.

1.1 Why the plane is easy

Multiplying a complex number by $e^{i\theta}$ turns it through the angle θ and leaves its length alone. Most of what makes complex numbers useful in physics follows from that one fact: phases, waves, the rotation group of the plane written as a single exponential. It is natural to ask for the same thing in three dimensions, and William Rowan Hamilton spent more than a decade trying to build it out of triples $a + bi + cj$.

He could not, and there is a good reason. Rotations in space do not commute: turn a book a quarter turn about one axis and then about another, and you get a different result from doing it the other way round. A number system that rotates space must therefore have a non-commutative product. In October 1843 Hamilton saw that this can be done if one allows a fourth unit, and he scratched the rule into a bridge in Dublin:

$$i^2 = j^2 = k^2 = ijk = -1. \tag{1.1}$$

1.2 The algebra

A *quaternion* is an expression $q = a+bi+cj+dk$ with real a, b, c, d. Addition is componentwise. Multiplication is fixed by (1.1) and the distributive law, and from (1.1) alone one finds

$$ij = k, \quad jk = i, \quad ki = j, \qquad ji = -k, \quad kj = -i, \quad ik = -j.$$

The products go round the cycle $i \to j \to k \to i$ with a plus sign and the other way with a minus sign. You will meet this cycle again in Chapter 3, where it becomes the defining feature of the operators that give the theory its name.

Lean checks the four relations of (1.1) directly in Mathlib's quaternions (`i_sq`, `j_sq`, `k_sq`, `ijk`). Each proof is a line of arithmetic, which is reassuring: the whole edifice rests on four lines that nobody needs to take on trust.

It helps to split a quaternion into a real part and a vector part, $q = a + \mathbf{u}$ with $\mathbf{u} = bi + cj + dk$. For two pure quaternions the product is

$$\mathbf{u}\,\mathbf{v} = -\,\mathbf{u}\cdot\mathbf{v} + \mathbf{u}\times\mathbf{v}. \tag{1.2}$$

So the quaternion product already contains the dot product and the cross product, glued together. Historically this is where vector analysis came from; Gibbs and Heaviside later pulled the two pieces apart.

1.3 Conjugate, norm and inverse

The *conjugate* of $q = a + \mathbf{u}$ is $\bar{q} = a - \mathbf{u}$, and one checks that $q\bar{q} = \bar{q}q = a^2 + b^2 + c^2 + d^2$. We write this as $|q|^2$. The crucial property is that the norm is multiplicative.

Theorem 1.1. proved in Lean *For all quaternions p, q, $|pq|^2 = |p|^2\,|q|^2$.*

In the repository this is `normSq_mul'`.

Written out in components, the theorem says that a product of two sums of four squares is again a sum of four squares, an identity Euler had found a century before Hamilton. It also means that every nonzero quaternion has an inverse, $q^{-1} = \bar{q}/|q|^2$, so the quaternions form a division algebra: you can divide, as long as you remember on which side.

1.4 Quaternions as matrices

There is a second way to look at all of this, and it is the one physicists use most. Send the quaternion $q = a + bi + cj + dk$ to the complex matrix

$$\rho(q) = \begin{pmatrix} a + ib & c + id \\ -c + id & a - ib \end{pmatrix}. \tag{1.3}$$

Proposition 1.2. proved in Lean *The map ρ respects products, $\rho(pq) = \rho(p)\rho(q)$, it sends the conjugate to the Hermitian adjoint, $\rho(\bar{q}) = \rho(q)^\dagger$, it is injective, and*

$$\det \rho(q) = |q|^2.$$

Lean: `rho_mul`, `rho_star`, `rho_injective`, `det_rho`.

The determinant formula is the matrix version of the multiplicative norm: determinants multiply, and so do norms. Under ρ the units i, j, k become i times the Pauli matrices, up to ordering and sign. Quaternions and the spin matrices of quantum mechanics are the same object seen from two sides.

1.5 A second determinant

The matrix picture invites one more step. Take four real numbers (t, x, y, z) and form the Hermitian matrix

$$X = \begin{pmatrix} t + z & x - iy \\ x + iy & t - z \end{pmatrix}.$$

Its determinant is

$$\det X = t^2 - x^2 - y^2 - z^2,$$

which is the Minkowski interval of special relativity (`det_lorentz`). So the same 2×2 matrices carry two quadratic forms. When the matrix comes from a quaternion, the determinant is the

Euclidean length $a^2 + b^2 + c^2 + d^2$. When it is Hermitian, the determinant is the Lorentzian length $t^2 - x^2 - y^2 - z^2$.

This is the first appearance of an idea that runs through the whole book. The round sphere of Euclidean geometry and the hyperboloid of relativity are two real slices of one complex object. Chapter 4 makes this precise, and Chapter 19 shows that the dumbbell lives on both slices at once.

Where this stands

Everything in this chapter is classical and serves as the foundation. The only claim is that it has been checked: the four defining relations, the multiplicative norm, the matrix representation and both determinant formulas are theorems in the repository.

Exercises

Exercise 1.1. Use (1.1) to derive $ij = k$ and $ji = -k$. *Answer:* the relations are checked in `i_sq`, `j_sq`, `k_sq`, `ijk`.

Exercise 1.2. Write out $|pq|^2 = |p|^2|q|^2$ in components and recognise Euler's four-square identity. *Answer:* `normSq_mul'`.

Exercise 1.3. Compute $\det \rho(q)$ from (1.3). *Answer:* `det_rho`.

Exercise 1.4. Show that $\rho(q) = 0$ forces $q = 0$. *Answer:* `rho_injective`.

Exercise 1.5. Compute the determinant of the Hermitian matrix X above. *Answer:* `det_lorentz`.

Chapter 2

Rotations and the three-sphere

The question

How does a quaternion rotate a vector, and why does it take two quaternions to describe each rotation? The answer is a sandwich, $\mathbf{v} \mapsto q\mathbf{v}\bar{q}$, and the two quaternions are q and $-q$, which give the same sandwich. The set of all such q is a three-dimensional sphere, and that sphere will become the cap of the dumbbell.

2.1 The unit quaternions

A quaternion with $|q| = 1$ is a *unit quaternion*. Writing $q = a + bi + cj + dk$, the condition is $a^2+b^2+c^2+d^2 = 1$: the unit quaternions form the three-sphere S^3 sitting in four-dimensional space. Because the norm is multiplicative, the product of two unit quaternions is again a unit quaternion, and so is the inverse $\bar{q}$. The three-sphere is therefore not just a shape but a group.

Through the matrix map of Chapter 1 we can say which group.

Proposition 2.1. proved in Lean *If $|q| = 1$ then $\rho(q)$ is unitary with determinant one: $\rho(q)\rho(q)^\dagger = 1$ and $\det \rho(q) = 1$. In other words, unit quaternions are the elements of $SU(2)$.*

Lean: `rho_SU2`.

This is the identification $S^3 = SU(2)$ that every course on quantum spin uses, usually without saying so.

2.2 The sandwich

Now take a pure quaternion $\mathbf{v} = xi + yj + zk$, which we think of as an ordinary vector, and a unit quaternion q. The product $q\mathbf{v}\bar{q}$ has two properties that make it a rotation.

Proposition 2.2. proved in Lean *For all quaternions q and x,*

$$|qx\bar{q}|^2 = |q|^4\,|x|^2, \qquad \operatorname{Re}(qx\bar{q}) = |q|^2 \operatorname{Re} x.$$

Lean: `hopf_norm`, `hopf_re`.

When $|q| = 1$ the first identity says that lengths are kept, and the second says that a pure quaternion stays pure. So $\mathbf{v} \mapsto q\mathbf{v}\bar{q}$ is a length-preserving linear map of three-dimensional space. One checks that it also preserves orientation, so it is a rotation. If $q = \cos(\theta/2) + \sin(\theta/2)\,\mathbf{n}$ with a unit vector $\mathbf{n}$, the rotation is through the angle θ about $\mathbf{n}$.

Notice the half-angle. As θ runs from 0 to 2π, the quaternion only gets from 1 to -1. A full turn of space corresponds to half a trip round the sphere, and one needs a turn through 4π to

bring q back to itself. Since q and $-q$ produce the same sandwich, each rotation corresponds to exactly two unit quaternions. This is the double cover $SU(2) \to SO(3)$, and it is the reason electrons have spin one half: their states transform under q, not under the rotation it produces.

2.3 The geometry of S^3

It is worth getting a feeling for the three-sphere itself, because it is the spatial shape of the cap. Fix the real part $a = \cos\chi$ with $0 \le \chi \le \pi$. The remaining components then satisfy $b^2 + c^2 + d^2 = \sin^2\chi$, which is an ordinary two-sphere of radius $\sin\chi$. So S^3 is a stack of two-spheres that start as a point at $\chi = 0$, grow to radius one at the equator $\chi = \pi/2$, and shrink back to a point at $\chi = \pi$. Its metric is

$$d\chi^2 + \sin^2\chi\, d\Omega^2,$$

and in Chapter 16 the areal radius $r = \ell \sin(x/\ell)$ of the cap will be exactly this function with $x = \ell\chi$. The cap of the dumbbell is the northern half of this stack, cut at the equator, where the spheres are largest.

The map $q \mapsto q\, i\, \bar{q}$ goes further. It sends each unit quaternion to a unit pure quaternion, that is, to a point of the ordinary two-sphere, and the points that land on the same place form a circle. This is the Hopf fibration $S^3 \to S^2$ with circles as fibres. We will meet it again in Chapter 12 as the bundle of a magnetic monopole.

Where this stands

All of this is classical. The chapter's role is to fix two facts that are used everywhere later: unit quaternions are $SU(2)$, and S^3 is a stack of two-spheres whose radius is $\sin\chi$.

Exercises

Exercise 2.1. Show that $\rho(q)$ is unitary with determinant one when $|q| = 1$. *Answer:* `rho_SU2`.

Exercise 2.2. Show that $qx\bar{q}$ has real part $|q|^2 \operatorname{Re} x$. Why does this mean that pure quaternions are mapped to pure quaternions when $|q| = 1$? *Answer:* `hopf_re`.

Exercise 2.3. Show that $|qx\bar{q}|^2 = |q|^4|x|^2$. *Answer:* `hopf_norm`.

Exercise 2.4. Check that $\rho(1)$ is the identity matrix and $\rho(\bar{q}) = \rho(q)^\dagger$. *Answer:* `rho_one`, `rho_star`.

Chapter 3

Fueter operators

The question

Complex analysis begins with the Cauchy–Riemann operator $\partial_x + i\partial_y$, whose square root of the Laplacian makes holomorphic functions harmonic. Is there a quaternionic version, and how many are there? There are several, and the theory takes its name from one particular family of four: the *cyclic* Fueter operators. They are singled out by a simple requirement, that the four operators share a common solution.

3.1 Square roots of the Laplacian

In the plane, $(\partial_x - i\partial_y)(\partial_x + i\partial_y) = \partial_x^2 + \partial_y^2$. The Cauchy–Riemann operator is a square root of the Laplacian, and that is why the real and imaginary parts of a holomorphic function are harmonic.

Rudolf Fueter asked in the 1930s for the same thing in four dimensions with quaternion coefficients. With coordinates $x_0, \dots, x_3$ and $\partial_a = \partial/\partial x_a$, set

$$D = \partial_0 + i\partial_1 + j\partial_2 + k\partial_3, \qquad \bar{D} = \partial_0 - i\partial_1 - j\partial_2 - k\partial_3,$$

acting on quaternion-valued functions with coefficients multiplied from the left.

Theorem 3.1. proved in Lean *For commuting derivatives,*

$$\bar{D}D = D\bar{D} = \Delta_4 = \partial_0^2 + \partial_1^2 + \partial_2^2 + \partial_3^2,$$

while $D^2 = \Box + 2\partial_0(i\partial_1 + j\partial_2 + k\partial_3)$ *with* $\Box = \partial_0^2 - \partial_1^2 - \partial_2^2 - \partial_3^2$, *and* $D^2 + \bar{D}^2 = 2\Box$.

Lean: `conj_product`, `product_conj`, `square`, `square_sum`.

The last two formulas are worth a second look. The Euclidean Laplacian comes from $\bar{D}D$, but the wave operator of relativity is already hiding in D^2. If we allow a complex unit I that commutes with everything and replace the vector part by $I(i\partial_1 + j\partial_2 + k\partial_3)$, the hidden term disappears and $\bar{D}_I D_I = \Box$ exactly (`biquaternion_box`). This is the operator version of the observation at the end of Chapter 1: one algebra, two signatures.

3.2 Which coefficients work?

Any operator $\sum_a c_a \partial_a$ with quaternion coefficients can be tested in the same way. Its symbol is $\sum_a c_a \xi_a$, and the operator factorises the Laplacian when $\big(\sum_a \bar{c}_a \xi_a\big)\big(\sum_b c_b \xi_b\big) = |\xi|^2$ for every ξ.

Proposition 3.2. proved in Lean *The operator $\sum_a c_a \partial_a$ factorises the Laplacian in this sense if and only if*

$$\bar{c}_a c_b + \bar{c}_b c_a = 2\delta_{ab} \qquad \text{for all } a, b,$$

that is, if and only if (c_0, c_1, c_2, c_3) is an orthonormal basis of the quaternions.

Lean: `factorization_iff`.

So there is not one Fueter operator but many: one for every orthonormal frame. To make a definite choice one needs a further principle.

3.3 The cyclic family

Here is the principle used in this book. Instead of one operator, take a family of four, obtained from D by shifting the units round the cycle $1 \to i \to j \to k \to 1$:

$$D_m = \sum_{a=0}^{3} e_{m+a}\, \partial_a, \qquad (e_0, e_1, e_2, e_3) = (1, i, j, k),$$

with indices read modulo four. Each D_m factorises the Laplacian (`cyclic_factorizes`). The coefficients form the 4×4 quaternion matrix $C_{ma} = e_{m+a}$, and one can recover it from the operators themselves by letting them act on the coordinate functions (`coefficient_extraction`, `coefficient_unique`).

Why a cycle? Of the ways of shuffling four objects transitively with a group of order four, there are only two: the cyclic group and the Klein four-group. The two families behave very differently.

Theorem 3.3. proved in Lean *The cyclic coefficient matrix C is singular. Its right kernel is the line $\gamma(-k)\,\mathbb{H}$, where $\gamma(q) = (1, q, q^2, q^3)$, and its left kernel is $\mathbb{H}\,\gamma(i)$. Over the reals its image has dimension 12, so its quaternionic rank is 3. In contrast, the Klein family has trivial kernel.*

Lean: `right_kernel`, `left_kernel`, `rank_three`, `real_rank_twelve`, `klein_family_regular`.

The singularity has two readings, one for each kernel. The right kernel gives a common solution: for any constant quaternion q, the linear function $f = \sum_b x_b\, \gamma(-k)_b\, q$ is annihilated by all four operators at once (`null_solution`). The left kernel gives a relation among the operators themselves,

$$\sum_m \gamma(i)_m\, D_m f = 0 \qquad \text{for every } f,$$

so only three of the four are independent (`operator_relation`).

This is the precise sense in which the cyclic family is special: it is the minimal transitive family of Laplacian square roots that admits a common solution. Everything named "cyclic Fueter" in the rest of the book refers to it.

Where this stands

Fueter's operator and the factorisation of the Laplacian are classical. The derivation of the cyclic family from three requirements (factorise the Laplacian, permute the units transitively with the smallest group, share a common solution) and the computation of both kernels are part of the author's programme and are checked here. Whether the cyclic family has deeper meaning than this selection rule is an interpretation that the book does not rely on.

Exercises

Exercise 3.1. Verify $\bar{D}D = \Delta_4$ by expanding the product and using $i^2 = j^2 = k^2 = -1$ and the anticommutation of distinct units. *Answer:* `conj_product`.

Exercise 3.2. Show that $D^2 + \bar{D}^2 = 2\Box$. *Answer:* `square_sum`.

Exercise 3.3. Check that $\gamma(i) = (1, i, -1, -i)$ and $\gamma(-k) = (1, -k, -1, k)$. *Answer:* `gam_i`, `gam_mk`.

Exercise 3.4. Show that every row of C annihilates $\gamma(-k)$ from the right. *Answer:* `right_kernel_rows`.

Exercise 3.5. Show that the Klein family has no common solution. *Answer:* `klein_family_regular`.

Chapter 4

Lorentz transformations: rotation times boost

The question

Special relativity mixes space and time through boosts, while ordinary geometry only knows rotations. How are the two related? Every Lorentz transformation, written as a 2×2 complex matrix of determinant one, is a rotation followed by a boost, and in exactly one way. Rotations live on a three-sphere, boosts on a three-hyperboloid, and the two meet only at the identity.

4.1 Lorentz transformations as matrices

At the end of Chapter 1 we saw that a Hermitian 2×2 matrix X encodes a spacetime point (t, x, y, z) with $\det X = t^2 - x^2 - y^2 - z^2$. Take any complex matrix A with $\det A = 1$ and form $AXA^\dagger$.

Proposition 4.1. proved in Lean *If* $\det A = 1$ *and* X *is Hermitian, then* $AXA^\dagger$ *is Hermitian and* $\det(AXA^\dagger) = \det X$.

Lean: `lorentz_action`.

So every A in $SL(2, \mathbb{C})$ moves spacetime points while keeping the interval: it is a Lorentz transformation. As with rotations, A and $-A$ do the same thing, so $SL(2, \mathbb{C})$ is the double cover of the Lorentz group, exactly as $SU(2)$ was the double cover of the rotations.

4.2 Polar decomposition

Inside $SL(2, \mathbb{C})$ there are two kinds of special matrices. Unitary ones, $U^\dagger U = 1$, are the rotations of Chapter 2. Positive Hermitian ones, $B = B^\dagger$ with positive eigenvalues, are pure boosts: they stretch time along one direction and leave the perpendicular directions alone.

Theorem 4.2 (polar decomposition)**.** proved in Lean *Every* $M \in SL(2, \mathbb{C})$ *can be written as* $M = UB$ *with* U *unitary and* B *positive Hermitian, both of determinant one, and the two factors are unique.*

Lean: `polar_exists`, `polar_unique`.

The proof is pleasant because everything is 2×2. The matrix $M^\dagger M$ is Hermitian with determinant one (`gram_herm`, `det_gram`). For 2×2 matrices the Cayley–Hamilton theorem says

$X^2 = (\operatorname{tr} X)X - (\det X)\,1$ (`ch2`), and from it one can write down the square root explicitly:

$$B = \frac{M^\dagger M + 1}{\sqrt{\operatorname{tr}(M^\dagger M) + 2}},$$

which squares to $M^\dagger M$ (`boost_from_square`). Then $U = MB^{-1}$ is unitary. No spectral theorem is needed.

Proposition 4.3. *proved in Lean* *A matrix in $SL(2,\mathbb{C})$ that is both unitary and Hermitian with positive eigenvalues is the identity.*

Lean: `sphere_meets_hyperboloid`.

In geometric terms the rotations form a three-sphere, the boosts form a three-dimensional hyperboloid (the space of velocities), and the two touch at one point.

4.3 One matrix, two slices

There is a single formula behind all of this. For four complex numbers $z_0, \dots, z_3$ put

$$M(z) = z_0 + i\,(z_1\sigma_1 + z_2\sigma_2 + z_3\sigma_3), \qquad \det M(z) = z_0^2 + z_1^2 + z_2^2 + z_3^2.$$

When all z's are real, M is a multiple of a unitary matrix and the determinant is the Euclidean length: this is the three-sphere of rotations (`real_slice_unitary`). When z_0 is real and z_1, z_2, z_3 are imaginary, M is Hermitian and the determinant is the Minkowski length: this is the hyperboloid (`imag_slice_hermitian`). The step between them is multiplication of three components by i, the familiar Wick rotation (`wick_path`). It turns curvature $+1/R^2$ into $-1/R^2$ (`curvature_flip`).

Chapter 6 will show this in a second form. The hyperboloid of velocities can be drawn as a hemisphere, with the equator at infinite speed (`hemisphere_on_sphere`, `hemisphere_conformal`, `equator_at_infinity`). The quantum sphere of that chapter is exactly this hemisphere.

Why this matters for the dumbbell

The cap of the dumbbell is a round three-sphere (rotations) and the neck is $\mathrm{AdS}_2 \times S^2$, a space that is round in two directions and hyperbolic in the other two (rotation $\times$ boost). Chapter 19 shows that the self-dual connection $\Gamma + iK$ of general relativity splits in the same way, into a rotation part and a boost part.

Where this stands

The polar decomposition is classical. What is particular here is the explicit 2×2 proof without the spectral theorem, and the organisation of the sphere and the hyperboloid as the real and imaginary slices of one determinant, which the book uses repeatedly.

Exercises

Exercise 4.1. Prove the Cayley–Hamilton identity $X^2 = (\operatorname{tr} X)X - (\det X)1$ for 2×2 matrices. *Answer:* `ch2`.

Exercise 4.2. Show that $\det(M^\dagger M) = 1$ when $\det M = 1$. *Answer:* `det_gram`.

Exercise 4.3. Check that $B = (M^\dagger M + 1)/\sqrt{\operatorname{tr}(M^\dagger M) + 2}$ squares to $M^\dagger M$. *Answer:* `boost_from_square`.

Exercise 4.4. Compute $\det M(z)$ for the matrix above. *Answer:* `det_M`.

Exercise 4.5. Show that $1/\cosh\eta \le 2/(\eta+1)$ for $\eta \ge 0$, and explain why it puts the equator of the hemisphere at infinite rapidity. *Answer:* `equator_at_infinity`.

Part II

The quantum sphere and the mass shell

Chapter 5

The mass-shell hyperboloid

The question

What shape does relativistic momentum have? For a particle of mass m the energy and momentum satisfy $E^2 - p^2 = m^2$, which is a hyperboloid. This chapter learns to measure on it, with rapidity as the natural coordinate, and meets the Gudermann function that will reappear in the quantum sphere and on the neck of the dumbbell.

5.1 Velocity space

A particle of mass $m > 0$ and momentum $\mathbf{p}$ has energy $E = \sqrt{m^2 + |\mathbf{p}|^2}$. Divide by m and write $X_0 = E/m$ and $\mathbf{u} = \mathbf{p}/m$; then

$$X_0^2 - |\mathbf{u}|^2 = 1, \qquad X_0 > 0.$$

This is one sheet of a hyperboloid in four dimensions. Its points are four-velocities, so this three-dimensional surface is the space of all possible states of motion of the particle. It is the boost half of Chapter 4: every point is reached from the rest point $(1, \mathbf{0})$ by exactly one boost.

The natural coordinate along a radial line is the *rapidity* η, defined by $|\mathbf{p}| = m \sinh \eta$ and $E = m \cosh \eta$. Velocities do not add simply, but rapidities along one line do, which is why η rather than v measures distance on the hyperboloid.

5.2 The geometry of the hyperboloid

With the metric induced from Minkowski space, the hyperboloid of radius m has

$$ds_H^2 = m^2 \big(d\eta^2 + \sinh^2 \eta \, d\Omega^2 \big),$$

which is hyperbolic space of curvature $-1/m^2$. A short computation along a radial line shows what this means in practice.

Proposition 5.1. proved in Lean *On the mass shell $E(p) = \sqrt{m^2 + p^2}$ the slope is $dE/dp = p/E = v$, the velocity. The outward normal in the Minkowski sense is $(E/m, p/m)$, the four-velocity, and it has Minkowski length -1. At $p = 0$ the second derivative of $X_0(u) = \sqrt{1 + u^2}$ is $+1$.*

In the repository: `slope_H`, `normal_H`, and the curvature at the contact point inside `cayley_klein`.

So the shell bends upward away from the rest point, and the normal to the shell at any point is the four-velocity of the particle in that state. Integrating, a disc of rapidity radius η_0

in a two-dimensional slice of the shell has area

$$2\pi m^2(\cosh\eta_0 - 1) = 2\pi m\,(E_0 - m),$$

proportional to the kinetic energy at its rim (`area_H`). Its curvature and the geodesic curvature of its boundary obey the Gauss–Bonnet theorem, as they must (`gauss_bonnet_H`).

5.3 The Gudermann function

One function connects rapidity to angles, and it will turn up again and again. It is the Gudermann function

$$\operatorname{gd}\eta = \arctan(\sinh\eta), \qquad \frac{d}{d\eta}\operatorname{gd}\eta = \frac{1}{\cosh\eta}.$$

Proposition 5.2. proved in Lean *If $\chi = \operatorname{gd}\eta$, then*

$$\sin\chi = \tanh\eta = v, \qquad \cos\chi = \frac{1}{\cosh\eta} = \frac{1}{\gamma}, \qquad \tan\chi = \sinh\eta = \frac{p}{m}.$$

The function gd *is strictly increasing and maps the real line onto $(-\pi/2, \pi/2)$.*

In the repository: `gudermann`, `gudermann_momentum`, `gd_deriv`, `gd_strictMono`, `gd_surj`.

In words: an angle χ that runs only up to a right angle can record every rapidity from zero to infinity. The cosine of that angle is the inverse Lorentz factor and the sine is the velocity. Infinite speed sits at $\chi = \pi/2$, at a finite angle. Chapter 6 turns this into a picture, and in Chapter 17 the same function will describe the arrival of a wave front on the neck of the dumbbell.

Where this stands

Classical. The chapter's job is to fix rapidity and the Gudermann function as the coordinates in which the quantum sphere and the mass shell will be compared.

Exercises

Exercise 5.1. Show that $dE/dp = p/E$. *Answer:* `slope_H`.

Exercise 5.2. Show that the Minkowski normal of the shell is the four-velocity and has length -1. *Answer:* `normal_H`.

Exercise 5.3. Prove $\sin(\operatorname{gd}\eta) = \tanh\eta$ and $\cos(\operatorname{gd}\eta) = 1/\cosh\eta$. *Answer:* `gudermann`.

Exercise 5.4. Differentiate gd. *Answer:* `gd_deriv`.

Exercise 5.5. Compute the area of a disc of rapidity radius η_0 on the shell. *Answer:* `area_H`.

Chapter 6

The quantum sphere

The question

Quantum mechanics describes a free relativistic particle by a state that depends on its momentum. Where do these states live? Normalised, they fill a hemisphere of a three-sphere, the *quantum sphere*. It touches the mass shell of Chapter 5 at the rest point and curves the opposite way, and its height is the reciprocal of the shell's height.

6.1 The map Q

For $m > 0$ and $\mathbf{p} \in \mathbb{R}^3$ put

$$Q(\mathbf{p}) = \Big(\frac{m}{E}, \frac{\mathbf{p}}{E}\Big), \qquad E = \sqrt{m^2 + |\mathbf{p}|^2}.$$

The four numbers satisfy $(m/E)^2 + |\mathbf{p}/E|^2 = 1$, so $Q(\mathbf{p})$ is a unit quaternion, a point of S^3, and its first component is positive (`qD_unit`, `qD_upper`). In matrix language it is the unitary matrix $U(\mathbf{p}) = (m - i\boldsymbol{\sigma}\cdot\mathbf{p})/E$, which appears in the free Dirac equation; Chapter 7 explains why.

Theorem 6.1. proved in Lean *Q is a bijection from $\mathbb{R}^3$ onto the open upper hemisphere $\{(w, \mathbf{x}) : w > 0,\ w^2 + |\mathbf{x}|^2 = 1\}$, with inverse $\mathbf{p} = m\mathbf{x}/w$.*

In the repository: `qmap_bij`, with `qmap_inj`, `qmap_surj`, `qinv_left`, `qinv_right`.

Every momentum gives a point of the hemisphere, and every point is reached once. Rest is the north pole. Momenta of ever larger size crowd towards the equator, which they never reach: the equator is infinite momentum.

6.2 Reciprocal heights

Put the quantum sphere and the mass shell side by side, in the coordinates of Chapter 5. The shell point is $(X_0, \mathbf{u}) = (E/m, \mathbf{p}/m)$ and the sphere point is $(w, \mathbf{x}) = (m/E, \mathbf{p}/E)$.

Proposition 6.2. proved in Lean *$wX_0 = 1$ and $\mathbf{x} = \mathbf{u}/X_0$. The polar angle of Q is the Gudermann angle of the rapidity, $\chi = \operatorname{gd}\eta$.*

In the repository: `qh_reciprocal`, `gudermann`.

So the two surfaces are mirror images through the plane of Newtonian physics. Both pass through the rest point with zero slope, the shell bends up with second derivative $+1$ and the sphere bends down with -1. They belong to a single family $X_0 = \sqrt{1 - \kappa X^2}$, all tangent at the rest point, with second derivative $-\kappa$: the sphere is $\kappa = 1$, the shell $\kappa = -1$, and Newton is $\kappa = 0$ (`cayley_klein`). Restoring units, $\kappa = \pm 1/c^2$.

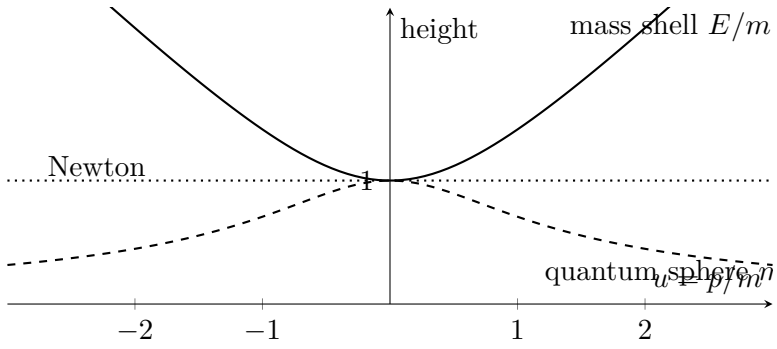


Figure 6.1: The mass shell (solid) and the quantum sphere (dashed) over the same proper velocity u. They touch at rest with the Newtonian plane (dotted) and bend in opposite directions; their heights are reciprocal.

6.3 The metric of the quantum sphere

The quantum sphere is not just a picture; it has a metric, induced from S^3, and it is related to the hyperbolic metric of the shell by a simple factor.

Proposition 6.3. proved in Lean *The metric of S^3 pulled back by Q is*

$$\partial_i Q \cdot \partial_j Q = \frac{\delta_{ij}}{E^2} - \frac{p_i p_j}{E^4},$$

and it equals the metric of the mass shell divided by E^2: the quantum hemisphere is conformal to hyperbolic space.

In the repository: `pullback_metric`, `conformal`, `metric_pullback`.

The conformal factor has a direct consequence for areas. A cap of the hemisphere of polar radius $\chi_0 = \operatorname{gd} \eta_0$ has area $2\pi(1 - m/E_0)$, and the corresponding disc on the shell has area mE_0 times larger (`area_Q`, `area_ratio`). As $E_0 \to \infty$ the hemisphere's area tends to the finite value 2π while the shell's grows without bound. Infinite momentum space has been folded into a finite, round region, and it is round in the way that rotations are round.

A thought to carry forward

In Chapter 4 the sphere of rotations and the hyperboloid of boosts were the real and imaginary slices of one complex quadric. Here the same pair appears in physics: the hyperboloid is where the particle's motion lives, and the hemisphere is where its quantum state lives. The cap of the dumbbell will be a round S^3, and the neck will contain a hyperbolic plane.

Where this stands

The map Q and its pullback metric are elementary, and the relation to the Dirac Hamiltonian is standard. The organisation of shell, sphere and Newtonian plane into one family tangent at rest, and the bookkeeping of areas, are presentations developed for this book; they are checked but not claimed as research results.

Exercises

Exercise 6.1. Show that $Q(\mathbf{p})$ is a unit quaternion with positive real part. *Answer:* `qD_unit`, `qD_upper`.

Exercise 6.2. Show that Q is injective. *Answer:* `qmap_inj`.

Exercise 6.3. Show that $wX_0 = 1$. *Answer:* `qh_reciprocal`.

Exercise 6.4. Compute the second derivatives of $\sqrt{1+u^2}$ and $1/\sqrt{1+u^2}$ at $u = 0$. *Answer:* `curv_H_Q`, `cayley_klein`.

Exercise 6.5. Compute the area of a cap of the hemisphere. *Answer:* `area_Q`.

Chapter 7

The Dirac equation and its projector

The question

Why does the unit quaternion $U(\mathbf{p}) = (m - i\boldsymbol{\sigma} \cdot \mathbf{p})/E$ of Chapter 6 deserve the name "quantum"? Because it is what the Dirac equation produces when one asks for the states of positive energy. The projector onto those states has a natural metric, and that metric is exactly half the metric of the quantum sphere.

7.1 A square root of E^2

Dirac looked for a Hamiltonian linear in momentum whose square is $E^2 = m^2 + |\mathbf{p}|^2$. Writing $H = m\beta + \alpha_1 p_1 + \alpha_2 p_2 + \alpha_3 p_3$ and squaring, the cross terms must cancel and the squares must be one.

Proposition 7.1. proved in Lean *If $\beta^2 = \alpha_i^2 = 1$ and $\beta, \alpha_1, \alpha_2, \alpha_3$ anticommute pairwise, then*

$$H^2 = (m^2 + p_1^2 + p_2^2 + p_3^2)\, 1.$$

In the repository: `dirac_sq`. The proof holds in any associative algebra, so it does not care which matrices one uses.

Compare this with Chapter 3. There we wanted a first-order operator whose "square" is the Laplacian, and quaternion units did the job. Here we want one whose square is E^2, and anticommuting matrices do the job. The two problems are the same problem, and in the standard representation the α_i are built from the quaternion units of Chapter 1.

7.2 The positive-energy projector

Since $H^2 = E^2$, the eigenvalues of H are $\pm E$, and

$$P_+ = \tfrac{1}{2}\Big(1 + \frac{H}{E}\Big)$$

is the projector onto positive energy.

Proposition 7.2. proved in Lean *If $H^2 = E^2$ and $E \neq 0$, then $P_+^2 = P_+$.*

In the repository: `proj_idem`.

As $\mathbf{p}$ varies, $P_+(\mathbf{p})$ traces out a family of two-dimensional subspaces, and any such family carries a metric, the *quantum metric* $\frac{1}{2}\operatorname{tr}(\partial_i P_+\, \partial_j P_+)$. It measures how fast the positive-energy states change with momentum.

Theorem 7.3. proved in Lean *The quantum metric of P_+ equals one half of the metric pulled back by the map Q of Chapter 6:*

$$\tfrac{1}{2}\operatorname{tr}(\partial_i P_+\,\partial_j P_+) = \tfrac{1}{2}\Big(\frac{\delta_{ij}}{E^2} - \frac{p_i p_j}{E^4}\Big).$$

In the repository: `quantum_metric`, together with `pullback_metric`.

So the space of positive-energy Dirac states, measured in its own natural way, is a hemisphere of radius $1/\sqrt{2}$. Along a radial line the polar angle is $\chi = \arctan(p/m)$, whose derivative is m/E^2 (`chi_deriv`, `radial_metric`). The "quantum sphere" is therefore not a metaphor: it is the geometry of Dirac states.

7.3 The non-relativistic limit

The Dirac equation should reduce to the Schrödinger–Pauli equation at low speed. The standard way to see this is the Foldy–Wouthuysen transformation, which removes the terms that mix upper and lower components order by order in $1/m$. Write $H = m\beta + \mathcal{O} + \mathcal{E}$ with an odd part $\mathcal{O}$ (anticommuting with β) and an even part $\mathcal{E}$.

Theorem 7.4. proved in Lean *After one transformation with generator $\beta\mathcal{O}/(2m)$ the Hamiltonian is, through order $1/m^2$,*

$$\beta\Big(m + \frac{\mathcal{O}^2}{2m}\Big) + \mathcal{E} - \frac{1}{8m^2}[\mathcal{O},[\mathcal{O},\mathcal{E}]],$$

plus odd terms of order $1/m$. A second transformation removes those odd terms without changing the even part at this order.

In the repository: `fw_order2`, `fw_hadamard3`, `second_fw`, `odd1_factor`.

With $\mathcal{O} = \boldsymbol{\alpha}\cdot\boldsymbol{\pi}$ in an external electromagnetic field, $\mathcal{O}^2/2m$ becomes the Pauli Hamiltonian with its $g = 2$ magnetic moment, and the double commutator produces the Darwin term and the spin–orbit coupling (`pauli_nc`, `darwin_form`, `so_form`). The algebra is exact; the expansion itself is a formal series in $1/m$, which the source states explicitly.

Where this stands

The Dirac equation, the projector and Foldy–Wouthuysen are textbook material. The identification of the quantum metric with half the pullback metric of Q is the bridge that the rest of Part II stands on; it is a short computation, checked here, and the author has not found it stated in this form.

Exercises

Exercise 7.1. Square $m\beta+\boldsymbol{\alpha}\cdot\mathbf{p}$ using only the anticommutation relations. *Answer:* `dirac_sq`.

Exercise 7.2. Show that P_+ is idempotent. *Answer:* `proj_idem`.

Exercise 7.3. Show that $\big(-mp/E^3\big)^2 + \big(m^2/E^3\big)^2 = (m/E^2)^2$. What does it say about the radial direction? *Answer:* `radial_metric`.

Exercise 7.4. Differentiate $\arctan(p/m)$. *Answer:* `chi_deriv`.

Chapter 8

Connections and curvature: Fueter, Berry, Wigner

The question

Carry a spinning particle round a loop in momentum space and bring it back. Does its spin come back the same? Not necessarily, and there are three natural ways to say by how much: the Fueter frame, the Berry phase of quantum mechanics, and the Wigner rotation of relativity. Two of them turn out to be the same connection in different gauges. The third is genuinely different, and one number tells them apart.

8.1 What a connection is

A connection tells you how to compare things at nearby points. In our setting the "things" are two-component spinors attached to each momentum $\mathbf{p}$, and a connection is a matrix-valued one-form $A = \sum_i A_i\, dp_i$ with A_i in $\mathfrak{su}(2)$. Moving a spinor along a path with $\dot\psi = -A(\dot{\mathbf{p}})\psi$ is *parallel transport*. Its failure to return round a small loop is measured by the curvature $F = dA + A \wedge A$.

Two connections are *gauge equivalent* if they differ only by a change of basis at each point, $A' = gAg^{-1} + g\, d(g^{-1})$. Then $F' = gFg^{-1}$, so anything built from F that is unchanged by conjugation is a gauge invariant.

8.2 Three connections

Using $U(\mathbf{p}) = (m - i\boldsymbol{\sigma}\cdot\mathbf{p})/E$ from Chapter 6:

- the *Fueter* (frame) connection is $A_F = \frac{1}{2}U^{-1}dU$;
- the *Berry* connection of positive-energy Dirac spinors is $A_B = \dfrac{i\boldsymbol{\sigma}\cdot(\mathbf{p}\times d\mathbf{p})}{2E(E+m)}$;
- the *Wigner* connection, the rotation part of the standard boost $L(\mathbf{p}) = \big((E+m) + \boldsymbol{\sigma}\cdot\mathbf{p}\big)/\sqrt{2m(E+m)}$, is $A_W = -\dfrac{i\boldsymbol{\sigma}\cdot(\mathbf{p}\times d\mathbf{p})}{2m(E+m)}$.

One reads off at once that $A_B = -(m/E)A_W$. They point the same way but have different strengths, and the ratio depends on $\mathbf{p}$. That alone does not prove they are inequivalent, because gauge transformations can do surprising things. One needs an invariant.

The Wigner rotation deserves its own sentence. For a Lorentz transformation Λ, the matrix $W = L(\Lambda\mathbf{p})^{-1}\Lambda L(\mathbf{p})$ is a rotation: unitary with determinant one (`wigner_unitary`, `wigner_det`), and it composes correctly, $W(\Lambda_2\Lambda_1, \mathbf{p}) = W(\Lambda_2, \Lambda_1\mathbf{p})W(\Lambda_1, \mathbf{p})$ (`wigner_cocycle`). Along a path of pure boosts it is transported exactly by A_W (`transport`), so Thomas precession is the holonomy of this connection.

8.3 The invariant that separates them

For any connection of the form $A_i = a(s)\, i\boldsymbol{\sigma}\cdot(\mathbf{p}\times e_i) + b(s)\, i\sigma_i$ with $s = |\mathbf{p}|^2$, one can compute the curvature in closed form and then the trace-Gram invariant $\mathcal{G} = \sum_k -2\,\mathrm{tr}(F_k^* F_k^*)$, which is unchanged by gauge transformations (`curv_formula`, `gram_trace`, `gram_gauge`).

Theorem 8.1. proved in Lean *For the three connections above,*

$$\mathcal{G}_F = \mathcal{G}_B = \Big(\frac{m}{E}\Big)^4 \mathcal{G}_W, \qquad \mathcal{G}_W = \frac{2}{m^2E^2} + \frac{1}{m^4}.$$

Hence for $E > m > 0$, that is for every nonzero momentum, $\mathcal{G}_B < \mathcal{G}_W$, and the Berry and Wigner connections are not gauge equivalent. The Fueter and Berry connections are gauge equivalent, with the gauge transformation given by the Foldy–Wouthuysen half rotation $V = \big((E+m) - i\boldsymbol{\sigma}\cdot\mathbf{p}\big)/\sqrt{2E(E+m)}$, which satisfies $V^2 = U$.

In the repository: `trG_W`, `trG_B`, `trG_F`, `ratio`, `berry_not_gauge_wigner`, `half_rotation_sq`, `berry_fueter_gauge`.

The pleasant surprise is the last sentence. The connection defined purely from the quaternion U (Fueter) and the one defined from quantum states (Berry) are the same object, and the change of gauge between them is the transformation that separates particles from antiparticles at low energy. The relativistic Wigner rotation is the odd one out, by a factor $(m/E)^4$ in the invariant. In the half-angle form, $A_B = \frac{1}{2}(1 - m/E)\, i\boldsymbol{\sigma}\cdot(\mathbf{n}\times d\mathbf{n})$ and $A_W = -\frac{1}{2}(E/m - 1)\, i\boldsymbol{\sigma}\cdot(\mathbf{n}\times d\mathbf{n})$ with $\mathbf{p} = |\mathbf{p}|\mathbf{n}$ (`berry_half_angle`, `wigner_half_angle`).

Where this stands

Berry phases of Dirac spinors and Thomas–Wigner rotation are each well known, and the relation $A_B = -(m/E)A_W$ is easy. The gauge-invariant proof that they are inequivalent, and the identification of Berry with Fueter through the Foldy–Wouthuysen half rotation, are results of the author's programme; they are checked here and the author has not found them in this form. A separate paper on the observable consequences of the difference between A_B and A_W is in preparation.

Exercises

Exercise 8.1. Show that the Wigner rotation is unitary with determinant one. *Answer:* `wigner_unitary`, `wigner_det`.

Exercise 8.2. Verify the cocycle rule. *Answer:* `wigner_cocycle`.

Exercise 8.3. Show that $\mathcal{G}$ is unchanged when F is replaced by gFg^{-1}. *Answer:* `gram_gauge`.

Exercise 8.4. Show that $V^2 = U$. *Answer:* `half_rotation_sq`.

Exercise 8.5. Show that $\mathcal{G}_B < \mathcal{G}_W$ whenever $E > m > 0$. *Answer:* `trG_B_lt`.

Chapter 9

Monopoles and zero modes

The question

A magnetic monopole is the simplest place where geometry forces a quantum state to exist. Put a charged spinning particle on a sphere around a monopole of charge q, and there are exactly $2q$ states of zero energy, all of one handedness. In this chapter the monopole appears twice: once in momentum space, where the Berry connection of Chapter 8 is a monopole, and once on an ordinary sphere, which will later be the cross-section of the neck.

9.1 A monopole in momentum space

The Berry connection of Chapter 8 lives on momentum space, but Chapter 6 taught us that momentum space carries a natural curved metric. Take the metric $h = (\delta_{ij} - p_i p_j/E^2)/m^2$. Along a radial line its length element is dp/E, and across it the length element is p/m times the round one (`h_radial`, `h_tangential`). Writing $p = m\sinh\eta$ turns this into

$$h = d\eta^2 + \sinh^2\eta\, d\Omega^2,$$

hyperbolic space with curvature exactly -1 (`h_rapidity`, `h3_curvature`).

Theorem 9.1. *proved in Lean* *On this hyperbolic space the Berry connection A_B and the Higgs field $\Phi = (\mathbf{p}/E)\cdot T$, with $T = -i\boldsymbol{\sigma}/2$, satisfy the Bogomolny equation*

$$D_A\Phi = -{*}F.$$

In the repository: `starF_closed` and its supporting lemmas in `AlphaCompleteC`.

So the Berry phase of a relativistic electron is not just any connection: it is a BPS monopole on the space of velocities. Its Higgs field is the velocity $\mathbf{p}/E$ itself, which goes from zero at rest to unit length at infinite speed, the equator of the quantum sphere. We will meet the Bogomolny equation again in Chapter 16, where an inequality of exactly this type governs the shape of the dumbbell.

9.2 Zero modes on a sphere

Now put a particle of charge q on a unit two-sphere around a monopole. After separating variables, the zero-energy condition for the Dirac operator reduces to a first-order equation in the polar angle,

$$u' = V(\theta)\,u, \qquad V = \frac{m + q\cos\theta}{\sin\theta},$$

with m the azimuthal quantum number. It can be solved exactly.

Proposition 9.2. proved in Lean *The solution is* $u(\theta) = \tan^m(\theta/2)\,\sin^q\theta$. *Near* $\theta = 0$ *it behaves like* θ^{m+q} *and near* $\theta = \pi$ *like* $(\pi-\theta)^{q-m}$, *so it is square integrable exactly when* $|m| < q + \frac{1}{2}$.

In the repository: `zero_mode`, `exponent_condition`, `zero_mode_L2`.

Counting the allowed values of m gives the index.

Theorem 9.3. proved in Lean *If* q *is an integer,* m *runs over* $2q$ *half-integers; if* q *is a half-integer, over* $2q$ *integers. In both cases there are exactly* $2q$ *zero modes, and the other chirality has none.*

In the repository: `count_integer_charge`, `count_half_charge`, `other_chirality_empty`.

This is the Atiyah–Singer index theorem in its simplest instance, derived here by hand: the number of zero modes equals the magnetic flux, and they all have the same handedness. One also needs to know that the other solutions of the radial equation are really not normalisable, not just that their leading terms are not; that is a short estimate (`large_not_L2`).

Looking ahead

The neck of the dumbbell is $\mathrm{AdS}_2 \times S^2$, and the Yang–Mills field on it is exactly a monopole of charge $1/2$ on the S^2 for each isospin component. By Theorem 9.3 each component carries one zero mode, which becomes a massless fermion in the two-dimensional AdS_2 (Chapter 25).

Where this stands

Monopole zero modes and their counting are classical. That the Berry connection of Dirac spinors, together with the velocity as a Higgs field, solves the Bogomolny equation on the space of velocities is part of the author's programme; it is checked here component by component, including the curvature -1 of the stage.

Exercises

Exercise 9.1. Show that the metric h becomes $d\eta^2 + \sinh^2\eta\, d\Omega^2$ under $p = m\sinh\eta$. *Answer:* `h_rapidity`.

Exercise 9.2. Check that $u = \tan^m(\theta/2)\sin^q\theta$ solves $u' = Vu$. *Answer:* `zero_mode`.

Exercise 9.3. Count the zero modes for $q = 1$ and for $q = 3/2$. *Answer:* `count_integer_charge`, `count_half_charge`.

Exercise 9.4. Show that a power θ^e is square integrable near 0 exactly when $e > -1/2$. *Answer:* `exponent_condition`.

Interlude A

The Standard Model from an internal sphere

The question

Can the quaternionic picture say anything about the particles we actually see? It can say something about their charges. A round nine-sphere has the symmetry $SO(10)$; stretching it into an ellipsoid breaks the symmetry to the Pati–Salam group, and the lowest Dirac modes carry exactly the charges of one generation of quarks and leptons. The dynamics that would select this ellipsoid is not derived, and this interlude says so clearly.

This interlude can be skipped without loss. It is placed here because it uses only the Dirac operator and the monopole of the preceding chapters, and because it is the part of the programme that touches particle physics most directly.

A.1 Why one $SU(2)$ is not enough

The quantum sphere of Chapter 6 is a copy of $SU(2)$. Electroweak physics needs $SU(2)_L \times U(1)_Y$, where the hypercharge Y commutes with all of $SU(2)_L$. Inside a single $\mathfrak{su}(2)$ that is impossible: a pure quaternion commuting with both i and j is zero (`center_pure_trivial`). With no Y the weak mixing angle would satisfy $\sin^2\theta_W = 1$ (`no_mixing`). This is a genuine obstruction, and the programme records it as a negative result. negative result

The way out is to use both sides. Left and right multiplication of quaternions commute with each other (`left_right_commute`), giving $SU(2)_L \times SU(2)_R$, and the inversion $U \mapsto U^{-1}$, which is momentum reversal $U(-\mathbf{p}) = U(\mathbf{p})^{-1}$, exchanges them (`inversion_swaps`, `U_parity`). With $Q = T_{3L} + T_{3R} + (B-L)/2$, one generation gets the right charges and $\sin^2\theta_W = 3/8$ at unification (`charges_leptons`, `charges_quarks`, `weinberg_full`).

A.2 The nine-sphere and Pati–Salam

Where do $B - L$ and colour come from? The round sphere $S^9 \subset \mathbb{R}^{10}$ has isometry group $SO(10)$. Split $\mathbb{R}^{10} = \mathbb{R}^4 \oplus \mathbb{R}^6$ and stretch the first factor by $\rho \neq 1$. Only matrices that do not mix the two factors survive (`ellipsoid_breaks`), leaving $SO(4) \times SO(6)$, locally $SU(2)_L \times SU(2)_R \times SU(4)_C$: the Pati–Salam group.

Theorem A.1. proved in Lean *The sixteen weights of the spinor representation of $SO(10)$, split as three plus two, carry the electric charge, $B - L$ and the two isospins of one Standard Model*

generation including a right-handed neutrino. The vector representation contains a Higgs bidoublet, and a Yukawa coupling $16 \cdot 16 \cdot 10$ *is allowed by the weights.*

In the repository: `spinor16_count`, `spinor16_charges`, `higgs_bidoublet`, `yukawa_weight`.

The direction of $B-L$ inside $SU(4)$ is fixed by colour symmetry and tracelessness, and its normalisation by the neutrality of the left-handed neutrino (`bl_direction`, `bl_normalization`). The $B-L$ anomaly cancels only if the right-handed neutrino is present (`bl_anomaly_cubic`, `bl_anomaly_needs_nuR`).

A.3 Chirality

The lowest Dirac modes of a round sphere are not zero modes: on S^9 the scalar curvature is 72, and the Lichnerowicz formula forces every harmonic spinor to vanish (`s9_scalar`, `no_zero_mode`). The index of the Dirac operator in odd dimensions is zero anyway (`index_zero`). So massless chiral fermions cannot come from the internal sphere alone. negative result

They can come from a domain wall. If a mass changes sign across a wall, there is exactly one normalisable zero mode, of one chirality (`jr_unique`, `kink_plus`, `norm_plus_lt`, `norm_minus`). This is the Jackiw–Rebbi mechanism, and Chapter 25 finds the same structure on the dumbbell: the mass W/r of the reduced fermion changes sign at the junction, and the zero mode sits there.

Where this stands

The group theory of $SO(10)$, Pati–Salam and the anomaly conditions are standard. What is specific here is the route: the quaternionic obstruction to $U(1)_Y$, its removal by left and right multiplication, and the ellipsoidal S^9 as the source of the breaking. The dynamics that would choose the ellipsoid, and the potential of the resulting Higgs field, are open; a curvature-only selection was tested and fails (Chapter 27). open

Exercises

Exercise A.1. Show that a pure quaternion commuting with i and j is zero. *Answer:* `center_pure_trivial`.

Exercise A.2. Show that a diagonal matrix commuting with the colour permutations has three equal entries. *Answer:* `bl_commute_color`.

Exercise A.3. Show that $\sum (B-L)^3 = 0$ over a generation with a right-handed neutrino and $\neq 0$ without it. *Answer:* `bl_anomaly_cubic`, `bl_anomaly_without_nuR`.

Exercise A.4. Show that $1/\cosh y$ is square integrable on the line and $\cosh y$ is not. *Answer:* `norm_plus_lt`, `norm_minus`.

Part III

Curved space and field equations

Chapter 10

Moving frames and curvature

The question

How do you compute the curvature of a curved space without drowning in indices? Cartan's answer is to attach an orthonormal frame to every point and watch how it turns. For the shapes in this book, a sphere-like cross-section of radius $r(x)$ swept along an axis, the whole computation fits on half a page, and the result will be used in every later chapter.

10.1 Frames and connection forms

On a curved space one cannot use Cartesian coordinates, but one can still choose, at every point, a set of orthonormal directions. Write the metric as $ds^2 = \sum_a (e^a)^2$ for one-forms e^a. How these frames rotate from point to point is recorded by the *connection forms* $\omega^a{}_b$, antisymmetric in a, b and fixed by Cartan's first structure equation

$$de^a + \omega^a{}_b \wedge e^b = 0.$$

The curvature is then the second structure equation, $\Omega^a{}_b = d\omega^a{}_b + \omega^a{}_c \wedge \omega^c{}_b$. Compared with Christoffel symbols this is simply a better way to organise the same information: the rotation of the frame is a small antisymmetric matrix, and the curvature measures how much that rotation fails to close round a loop.

10.2 The warped product

Every slice in this book has the form

$$ds^2 = dx^2 + f(x)^2\, d\Omega^2.$$

With the frame $e^x = dx$, $e^\theta = f\, d\theta$, $e^\phi = f \sin\theta\, d\phi$, the connection forms are

$$\omega^\theta{}_x = f'\, d\theta, \qquad \omega^\phi{}_x = f' \sin\theta\, d\phi, \qquad \omega^\phi{}_\theta = \cos\theta\, d\phi,$$

and these are torsion-free in all components (`spin_torsion_free`).

Theorem 10.1. proved in Lean *The sectional curvatures of $dx^2 + f^2 d\Omega^2$ are*

$$K_{x\theta} = K_{x\phi} = -\frac{f''}{f}, \qquad K_{\theta\phi} = \frac{1 - f'^2}{f^2},$$

and the scalar curvature is $R = \dfrac{2(1 - f'^2 - 2ff'')}{f^2}$.

In the repository: `warped_curvature`, `sectional`.

The two numbers have direct meaning. The "axial" curvature $-f''/f$ says how the cross-sections bend towards or away from each other along the axis. The "transverse" curvature $(1-f'^2)/f^2$ compares the cross-section with a flat cone.

Three examples will matter for the dumbbell.

- $f = \ell \sin(x/\ell)$: both curvatures are $1/\ell^2$, the round three-sphere of radius ℓ, which is the cap.
- $f = \ell$: the axial curvature is zero and the transverse one is $1/\ell^2$. This is the cylinder $\mathbb{R} \times S^2$, the spatial neck.
- $f = \sinh x$: both curvatures are -1, hyperbolic space, which we met as the space of velocities (`h3_curvature`).

10.3 Einstein's tensor for spherical spacetimes

The same method in four dimensions handles the two kinds of spherically symmetric spacetime metrics used later: the diagonal form $-N^2dt^2 + A^2dx^2 + R^2d\Omega^2$, and the double null form $-2\Omega\,du\,dv + r^2d\Omega^2$. The repository computes the Christoffel symbols directly from the Levi-Civita formula, then the Ricci and Einstein tensors, component by component (`einD_00`, `einD_11`, `einD_01`, `einN_00`, `einN_01`, `einN_22`). Two formulas from that computation will be quoted repeatedly:

$$G_{uu} = -\frac{2}{r}\Big(r_{uu} - \frac{\Omega_u}{\Omega} r_u\Big), \qquad G_{tt}/N^2 = \tfrac{1}{2}\,{}^3R \ \text{ at a moment of time symmetry.}$$

The first is the focusing of light, the second the Hamiltonian constraint. Neither is assumed later; both come out of this calculation.

10.4 Gauss–Bonnet

For a closed surface the integral of the Gaussian curvature is 2π times the Euler characteristic. On a round sphere of radius a this is 4π (`gauss_bonnet_round`); on a disc of the hyperbolic plane it balances the geodesic curvature of the boundary (`gauss_bonnet_H`). In Chapter 21 the same theorem turns the Euclidean action of the neck into a function of corner angles and topology only.

Where this stands

Standard differential geometry. What the repository adds is that every curvature formula used later is derived from the metric inside Lean rather than copied from a textbook.

Exercises

Exercise 10.1. Verify the first structure equation for the frame of $dx^2 + f^2d\Omega^2$. *Answer:* `spin_torsion_free`.

Exercise 10.2. Compute the sectional curvatures of $dx^2+f^2d\Omega^2$. *Answer:* `warped_curvature`.

Exercise 10.3. Show that $d\eta^2 + \sinh^2\eta\, d\Omega^2$ has curvature -1. *Answer:* `h3_curvature`.

Chapter 11

Slices and constraints

The question

In Newtonian mechanics you may choose the initial positions and velocities freely. In general relativity you may not: four of Einstein's ten equations contain no second time derivatives and must already hold on the initial slice. What do these constraints say, in plain words, for the slices of this book?

11.1 Initial data

A spacetime can be built by evolving a three-dimensional slice in time. The data on the slice are its metric h_{ij} and its extrinsic curvature K_{ij}, which says how the slice sits in spacetime, or equivalently how fast its metric is changing. The Einstein equations split into six evolution equations and four constraints. The constraints are

$$ {}^{3}R + K^2 - K_{ij}K^{ij} = 16\pi G\rho, \qquad D_j K^j{}_i - D_i K = 8\pi G\, j_i. $$

The first is the *Hamiltonian constraint*, the second the *momentum constraint*. At a moment of time symmetry, $K_{ij} = 0$, the momentum constraint is automatic and the Hamiltonian constraint says simply that the scalar curvature of space equals $16\pi G$ times the energy density.

11.2 Spherical slices

For $ds^2 = dx^2 + r(x)^2 d\Omega^2$ the scalar curvature is the one from Chapter 10, so at time symmetry

$$ \frac{2(1 - r'^2 - 2rr'')}{r^2} = 16\pi G\rho. \tag{11.1} $$

Away from time symmetry, with $K^i{}_j = \mathrm{diag}(a, b, b)$, the momentum constraint reduces to one equation (`codazzi_s`),

$$ \frac{2r'}{r}(a - b) - 2b' = 8\pi G\, j_x, $$

and its angular components vanish identically (`codazzi_theta`, `codazzi_phi`).

11.3 When the matter is Yang–Mills

Suppose the energy density is that of a magnetic Yang–Mills field. One can then ask a geometric question: is there a relation between the curvature of space and the energy density that does not depend on the particular solution? If the gauge field is identified with the

spin connection of the slice (we will see in Chapter 13 that this identification is natural), the Yang–Mills energy density becomes $|Rm|^2/(8g^2)$ and the Hamiltonian constraint reads

$$R = \frac{\ell^2}{2}\,|Rm|^2, \qquad \ell^2 = \frac{4\pi G}{g^2}$$

(`coeff_I`). In three dimensions the full curvature is determined by the Ricci tensor, $|Rm|^2 = 4|Ric|^2 - R^2$ (`rm2_3d`), so this is a condition on the Ricci tensor alone. It is the first time the length ℓ appears, and it appears as the ratio of Newton's constant to the gauge coupling.

Where this stands

The constraint equations are standard. The equation $R = (\ell^2/2)|Rm|^2$, obtained by identifying the gauge field with the spin connection, is specific to the programme, and Chapter 14 classifies its solutions.

Exercises

Exercise 11.1. Derive the reduced momentum constraint for $K = \mathrm{diag}(a, b, b)$. *Answer:* `codazzi_s`.

Exercise 11.2. Show that $|Rm|^2 = 4|Ric|^2 - R^2$ in three dimensions in terms of sectional curvatures. *Answer:* `rm2_3d`.

Chapter 12

Gauge fields

The question

Electromagnetism is the theory of a $U(1)$ connection; the strong and weak forces are theories of non-Abelian connections. What changes when the group stops commuting, and what does the simplest spherically symmetric $SU(2)$ field look like? The answer to the second question is a single function $W(r)$, and it will carry the whole matter content of the dumbbell.

12.1 Maxwell

Write the field strength as $F = dA$. Two sets of equations follow, and it is worth keeping them apart. The Bianchi identity $dF = 0$ is automatic, a consequence of $d^2 = 0$ (`bianchi`). The equation of motion $\partial_\mu F^{\mu\nu} = -J^\nu$ comes from the action, through the first variation (`first_variation`). Its time component is Gauss's law and its spatial components are Ampère's (`gauss`, `ampere`), and taking its divergence gives charge conservation (`current_conserved`, `charge_conservation`).

12.2 Yang–Mills

For $SU(2)$ the potential has three components A^a_μ and the field strength picks up a quadratic term,

$$F^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu + \varepsilon^{abc} A^b_\mu A^c_\nu .$$

The structure constants ε^{abc} are the commutators of the quaternion units of Chapter 1. Under a gauge transformation $A' = gAg^{-1} + g\,d(g^{-1})$ the field strength transforms by conjugation, $F' = gFg^{-1}$, in any associative algebra (`curvature_covariant`). The equation of motion is $D_\mu F^{a\mu\nu} = -J^{a\nu}$, and restricting to one colour direction gives back Maxwell (`ym_first_variation`, `u1_projection`).

12.3 Monopole bundles

A magnetic monopole of charge N cannot be described by one potential over the whole sphere; one needs two, on the northern and southern patches, related by a gauge transformation. Requiring that this transformation be single-valued forces Dirac's quantisation, and the integral of the curvature over the sphere, the Chern number, equals the charge (`dirac_quantization`, `chern_number`, `chern_eq_two_q`). The tangent bundle of S^2 is itself such a bundle with $|N| = 2$, which is the Euler characteristic seen as a magnetic charge (`AlphaVII_Tangent`).

12.4 The spherically symmetric $SU(2)$ field

The most general purely magnetic, spherically symmetric $SU(2)$ field is described by one function $W(r)$, in the form introduced by Witten. In an orthonormal frame its magnetic field has

$$B_r = \frac{1 - W^2}{r^2}, \qquad B_\theta = B_\phi = \frac{W'}{r}.$$

Two values of W are special. $W = \pm 1$ is pure gauge: no field at all. $W = 0$ is the embedded Abelian monopole of unit magnetic charge, a pure radial field $1/r^2$. Every configuration in this book interpolates between these.

The energy density of the field is $\rho = \frac{1}{2g^2}\big(2W'^2/r^2 + (1 - W^2)^2/r^4\big)$, and its stress is traceless, as for any Yang–Mills field in four dimensions. The tracelessness is what makes the cap a radiation universe in Chapter 15.

Where this stands

Standard gauge theory, checked in Lean, including the separation of the Bianchi identity from the field equations and the non-Abelian covariance.

Exercises

Exercise 12.1. Show that $F' = gFg^{-1}$ under a gauge transformation. *Answer:* `curvature_covariant`.

Exercise 12.2. Derive Gauss's law from the first variation of the Maxwell action. *Answer:* `gauss`.

Exercise 12.3. Show that the Chern number of a monopole of charge q is $2q$. *Answer:* `chern_eq_two_q`.

Chapter 13

The action principle

The question

Why this action and not another? For gravity coupled to an $SU(2)$ gauge field, a few natural requirements leave almost no freedom: the group is fixed by the quaternions, the gauge-field action by gauge invariance, the gravitational action by diffeomorphism invariance, and the boundary term by the requirement that the variational problem make sense. This chapter goes through each step and then checks that the reduced action reproduces the full four-dimensional equations.

13.1 The group

Start from the cyclic Fueter family of Chapter 3. Its common solutions form a right quaternionic line $\gamma(-k)\mathbb{H}$. Which real linear maps of a quaternionic line preserve both its right $\mathbb{H}$-structure and its norm? Exactly left multiplication by unit quaternions (`gauge_group`, `unit_left_mul`). So the internal symmetry at each point is $Sp(1) = SU(2)$, and its structure constants are the quaternion commutators (`su_two_bracket`).

13.2 The gauge-field action

Among Lorentz-invariant quadratic Lagrangians with two derivatives, the general symbol is $a(k \cdot k)(\xi \cdot \xi) + b(k \cdot \xi)^2 + m^2(\xi \cdot \xi)$. Gauge invariance $\xi \to \xi + tk$ forces $a + b = 0$ and $m = 0$ (`maxwell_symbol`, `maxwell_mass`), which is the Maxwell Lagrangian $-\frac{1}{4}F_{\mu\nu}F^{\mu\nu}$ up to normalisation. In the non-Abelian case the coefficient of the quadratic term in F is forced as well: covariance for all configurations requires it to be exactly one (`ym_coupling_forced`), and then $\sum_a F^a F^a$ is invariant (`ym_invariance`). The topological term εFF is a total derivative and does not affect the equations (`theta_total_derivative`).

13.3 The gravitational action

For a symmetric tensor field h, the Lorentz-invariant quadratic forms with two derivatives have four coefficients. Invariance under linearised diffeomorphisms $h \to h + k\xi + \xi k$ fixes them up to one overall constant (`fierz_pauli`). This is linearised Einstein–Hilbert. That its non-linear completion in four dimensions is unique, $R/(16\pi G)$ plus a cosmological constant, is Lovelock's theorem, which we quote. external theorem

13.4 The boundary term

The Einstein–Hilbert Lagrangian contains second derivatives of the metric. For a well-posed variational problem with the metric fixed on the boundary, those second derivatives must sit in a total derivative that a boundary term can cancel. For every spherically symmetric metric $-N^2dt^2 + A^2dx^2 + R^2d\Omega^2$ this is exactly what happens:

$$\frac{\sqrt{-g}\,R}{\sin\theta} = L_1 + \partial_t\big(2\sqrt{h}\,K_t\big) + \partial_x\big(-2\sqrt{-\gamma}\,K_x\big),$$

with L_1 containing first derivatives only (`ghy_spherical`). The Gibbons–Hawking–York term cancels the two total derivatives. So the action

$$S = \int \sqrt{-g}\Big[\frac{R}{16\pi G} - \frac{1}{4g^2}F^aF^a\Big] + \frac{1}{8\pi G}\oint K\sqrt{h}$$

is not chosen. It is what remains.

13.5 Reduction and the four-dimensional check

Reducing this action to spherical symmetry gives a two-dimensional action for N, A, R and W. Varying the reduced action is easier than varying the full one, but it raises a worry: are the reduced Euler–Lagrange equations really the same as the four-dimensional equations restricted to spherical symmetry? This is Palais' principle of symmetric criticality, which holds for compact symmetry groups but needs care.

The repository avoids the question by checking directly. It computes all sixteen components of the four-dimensional Einstein tensor and all twelve components of the Yang–Mills equations for the spherical ansatz and shows that they are equivalent to the reduced equations (`el_iff_alpha`, `match_N`, `match_W`, and the module `AlphaS2_4DMatch`).

Spin connection and gauge field

On the slice $dx^2 + r^2d\Omega^2$, the spin connection of Chapter 10 has components $r'd\theta$, $r'\sin\theta\,d\phi$, $\cos\theta\,d\phi$. These are precisely the Witten form of an $SU(2)$ field with $W = r'$ (`spin_is_witten`). This identification was the starting point of the dumbbell. Chapter 16 shows that, within conformally flat slices, it is not an assumption but the case of equality of an inequality.

13.6 Noether

One more consequence of having an action. If the Lagrangian is invariant under $\delta q = \varepsilon K(q)$, the quantity $\sum_i p_iK_i$ is conserved along solutions (`noether`). Cyclic coordinates are the special case $K = e_j$ (`cyclic_from_noether`). Every conservation law in later chapters comes from here.

Where this stands

Each step uses a standard result, and Lovelock's theorem is quoted. The direct check of all four-dimensional components, which removes the need for symmetric criticality, and the identification of the spin connection with the Witten ansatz are checked here.

Exercises

Exercise 13.1. Show that gauge invariance forces $a+b=0$ and $m^2=0$. *Answer:* `maxwell_mass`.

Exercise 13.2. Show that $\varepsilon^{\mu\nu\rho\sigma}F_{\mu\nu}F_{\rho\sigma}$ is a total derivative. *Answer:* `theta_total_derivative`.

Exercise 13.3. Find the coefficients of the Fierz–Pauli Lagrangian from gauge invariance. *Answer:* `fierz_pauli`.

Exercise 13.4. Prove Noether's theorem for a finite-dimensional Lagrangian. *Answer:* `noether`.

Part IV

The dumbbell

Chapter 14

Three phases

The question

Chapter 11 ended with an equation, $R = (\ell^2/2)|Rm|^2$, that a slice must satisfy when its gauge field is its own spin connection. Which shapes satisfy it? Only three: a round three-sphere of radius ℓ, flat space, and a cylinder of radius ℓ. These are the three building blocks from which every dumbbell is made, and no static dumbbell exists.

14.1 The candidates

Among the homogeneous three-geometries there are five natural candidates: flat space, the round sphere $S^3(a)$, hyperbolic space $H^3(a)$, and the two products $S^2(a) \times \mathbb{R}$ and $H^2(a) \times \mathbb{R}$. Testing each against $R = (\ell^2/2)|Rm|^2$ is a matter of plugging in constant curvatures.

Proposition 14.1. proved in Lean *The equation holds for flat space, for $S^3(\ell)$ and for $S^2(\ell) \times \mathbb{R}$. It fails for every hyperbolic space and for $H^2 \times \mathbb{R}$.*

In the repository: `satI_flat`, `satI_sphere`, `satI_sphProd`, `not_satI_hyp`, `not_satI_hypProd`, `three_phases`.

That these five are the only possibilities when the Ricci tensor is parallel is a classification theorem of Cheng, Ishikawa and Shiohama, which the repository quotes external theorem. For the spherically symmetric case, which is all the dumbbell needs, it proves the classification itself.

14.2 The spherical classification

For $ds^2 = dx^2 + r(x)^2 d\Omega^2$ with $W = r'$, the requirement that $W = r'$ survive the first moment of time evolution splits into cases (`sector_geodesic`, `phase_neck`, `phase_form`). Combined with the Hamiltonian constraint, each case has a clean global solution.

Theorem 14.2. proved in Lean *A spherically symmetric slice in one of the phases is, globally,*

(a) an S^3 phase, $r = r(0)\cos(x/\ell) + \ell r'(0)\sin(x/\ell)$ with $r(0)^2 + \ell^2 r'(0)^2 = \ell^2$, closing smoothly at its poles;

(b) a flat phase, $r = r(0) \pm x$;

(c) the neck, $r = \ell$.

In the repository: `s3_global`, `s3_amplitude`, `s3_pole_regular`, `flat_global`.

The proof of (a) is the energy method for the harmonic oscillator: the quantity $r^2+\ell^2 r'^2$ is conserved and fixes the amplitude to be ℓ. At the poles $r'^2 = 1$, which is exactly the condition for no conical singularity.

14.3 No static dumbbell

Can the pieces be glued into something static? Take the full static equations with a lapse $N(x)$, the Hamiltonian constraint, the radial Einstein equation and the Yang–Mills equation, all with $W = r'$.

Theorem 14.3. *proved in Lean* | *negative result* *On any interval, a static solution has either $W \equiv 0$ (and then $r \equiv \ell$, the neck) or $W \equiv \pm 1$ (and then the slice is flat). In particular no static solution joins a cap to a neck.*

In the repository: `static_W_trichotomy`, `static_classification`.

The proof deserves a word because it is unusual. On an open set where $W \neq 0$ and $W^2 \neq 1$, one differentiates the constraint three times along the flow and finds four polynomial expressions that must all vanish. A certificate, found by a Gröbner basis computation and checked in Lean, writes the nonzero quantity $\ell^2(rW(W^2-1))^4$ as a combination of those four (`cert`). So the open set is empty, and continuity does the rest.

Since the dumbbell cannot stand still, it must move. Chapters 15 and 18 describe how.

For the specialist

The general (non-spherical) problem of slices with harmonic curvature satisfying a quadratic Ricci constraint is treated in the repository through the class C2 of three distinct Ricci eigenvalues, with a two-stage compatibility computation (modules `C2`, `SS*`, `Aligned`). It shows that the aligned C2 case has no solutions. This material is not needed for the rest of the book.

Where this stands

The candidate list is standard and the general classification is quoted. The spherical classification, the global solutions and the non-existence of static dumbbells with a polynomial certificate are proved here.

Exercises

Exercise 14.1. Check that $S^3(\ell)$ satisfies $R = (\ell^2/2)|Rm|^2$ and $H^3(a)$ does not. *Answer:* `satI_sphere`, `not_satI_hyp`.

Exercise 14.2. Show that $r^2 + \ell^2 r'^2$ is constant when $r'' = -r/\ell^2$. *Answer:* `s3_amplitude`.

Exercise 14.3. Show that $r'^2 = 1$ where $r = 0$ on an S^3 phase. *Answer:* `s3_pole_regular`.

Chapter 15

Cap and neck

The question

What do the three phases become once they are allowed to move? The S^3 phase becomes a closed universe filled with Yang–Mills radiation that expands and recollapses. The cylinder becomes the throat of an extremal magnetic black hole, $\mathrm{AdS}_2 \times S^2$. Both have the size $\ell = \sqrt{4\pi G}/g$, for the same reason.

15.1 The stress of the field

For the Witten field of Chapter 12 the energy density and pressures are

$$\rho = \frac{1}{g^2}\Big(\frac{W'^2}{r^2} + \frac{(1-W^2)^2}{2r^4}\Big), \qquad p_r = \frac{1}{g^2}\Big(\frac{W'^2}{r^2} - \frac{(1-W^2)^2}{2r^4}\Big), \qquad p_T = \frac{(1-W^2)^2}{2g^2r^4},$$

and the trace $-\rho + p_r + 2p_T$ vanishes (`ym_stress`). On the two special shapes the stress becomes very simple (`cap_stress`, `neck_stress`).

- On the cap, $r = \ell\sin(x/\ell)$, $W = \cos(x/\ell)$: the stress is isotropic with $p = \rho/3$. It is radiation.
- On the neck, $r = \ell$, $W = 0$: $p_r = -\rho$ and $p_T = \rho$. It is the stress of a pure radial magnetic field, a tension along the field lines and a pressure across them.

15.2 One length for both

Theorem 15.1. proved in Lean *The Hamiltonian constraint holds on the cap if and only if $\ell^2 = 4\pi G/g^2$, and the same is true on the neck. On the neck all four components of the four-dimensional Einstein equations hold if and only if $\ell^2 = 4\pi G/g^2$.*

In the repository: `ell_cap`, `ell_neck`, `neck_einstein`.

So the cap and the neck do not merely happen to have the same size. The size is set by the ratio of gravitational to gauge coupling, and it is the same number on both. In Planck units $\ell = \sqrt{4\pi}/g$, a few Planck lengths for couplings of order one.

The magnetic flux through the spheres of the cap is $4\pi\sin^2(x/\ell)$; at the equator it is 4π, the flux through the neck (`flux_cap`). The field lines that enter the neck come from the cap; no magnetic charge is needed anywhere.

15.3 The cap moves: a closed radiation universe

A closed universe filled with radiation obeys the Friedmann equation. In conformal time η and with the radiation density $\rho = 3/(2g^2a^4)$ of the cap,

$$a'^2 + a^2 = \frac{8\pi G}{3}\rho a^4 = \ell^2,$$

solved by $a = \ell\cos\eta$ (`cap_friedmann_l`, `frw_exact`). The universe is momentarily static at $\eta = 0$, where its radius is ℓ, the neck's radius. It recollapses at $\eta = \pi/2$, at proper time ℓ for an observer at the centre, where the curvature diverges (Chapter 18). This is Tolman's radiation universe, here with the radiation being a homogeneous Yang–Mills field on $S^3 = SU(2)$.

15.4 The neck moves: $\mathrm{AdS}_2 \times S^2$

The neck is static in its own time. Its full spacetime is the product of a two-sphere of radius ℓ with two-dimensional anti-de Sitter space of radius ℓ, the Bertotti–Robinson solution. In static coordinates the lapse is $N = \cosh(x/\ell)$, which satisfies $N'' = N/\ell^2$ (`br_lapse_d1`, `br_lapse_d2`). The neck can carry the $W = r'$ structure with any lapse (`neck_any_lapse`), but it does not like perturbations of W: the mode W has negative mass squared on the neck (`neck_W_unstable`), a fact Chapter 20 makes precise.

Why the neck is a black-hole throat

Take a magnetically charged Reissner–Nordström black hole and push it to extremality, mass equal to charge in Planck units. The region near its horizon becomes an infinitely long tube with constant cross-section, and its geometry is exactly $\mathrm{AdS}_2 \times S^2$. The neck of the dumbbell is that tube, with the magnetic charge supplied by the $SU(2)$ field at $W = 0$.

15.5 Gluing

If the cap is glued to the neck at a sphere, continuity of the radius and of its first derivative already forces the cap's radius to equal ℓ (`glue_C1`), consistently with the constraints (`glue_EYM`). The neck cannot open into flat space (`no_flat_from_constraints`). Chapter 16 explains the structure of this gluing in full.

Where this stands

Both solutions are classical: Tolman's closed radiation universe and the Bertotti–Robinson spacetime. Their joint description, with one length fixed by G/g^2 for both and the magnetic flux passing from one to the other, is the starting point of the dumbbell programme.

Exercises

Exercise 15.1. Show that the Yang–Mills stress is traceless. *Answer:* `ym_stress`.

Exercise 15.2. Compute ρ, p_r and p_T on the cap and on the neck. *Answer:* `cap_stress`, `neck_stress`.

Exercise 15.3. Show that $a = \ell\cos\eta$ solves $a'^2 + a^2 = \ell^2$. *Answer:* `cap_friedmann_l`.

Exercise 15.4. Show that $\cosh(x/\ell)$ satisfies $N'' = N/\ell^2$. *Answer:* `br_lapse_d2`.

Chapter 16

The shape of the dumbbell

The question

Why can the cap and the neck not be joined smoothly? Because, as we will see, the neck is the envelope of the caps. Both solve the same first-order equation; the cap is its regular solution and the neck its singular one, and a curve that switches from a solution to the envelope of the solutions is C^1 but never C^2.

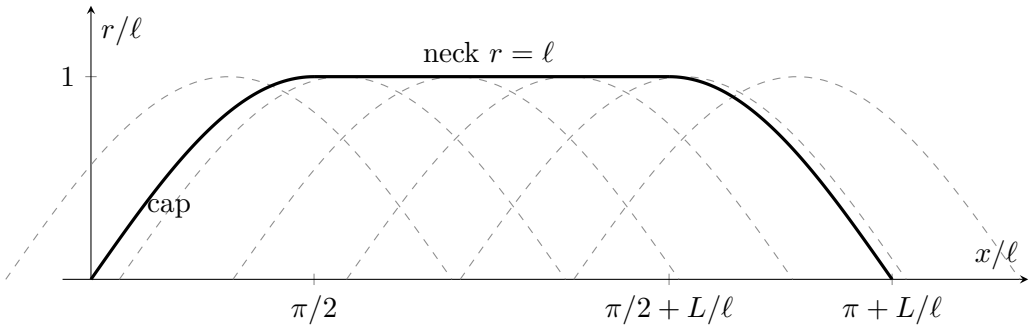


Figure 16.1: The areal radius along a time-symmetric slice of the dumbbell (solid). The dashed curves are other members of the family $r = \ell \sin((x - c)/\ell)$; the neck $r = \ell$ touches every one of them at its top. It is their envelope.

16.1 Kinematics: what a slice looks like

Freeze the dumbbell at a moment of time symmetry, when nothing is moving, and look at the space it occupies. Spherical symmetry lets us write its metric as

$$ds^2 = dx^2 + r(x)^2\, d\Omega^2, \tag{16.1}$$

where x is proper distance along the axis and $4\pi r^2$ is the area of the sphere at x; r is called the *areal radius*. The magnetic $SU(2)$ Yang–Mills field compatible with this symmetry is described by one function $W(x)$ (Chapter 12): its radial magnetic field is $(1 - W^2)/r^2$ and its tangential one is W'/r.

Two numbers attached to each of these spheres will carry most of the argument.

Definition 16.1. The *Misner–Sharp mass* of the sphere at x on a time-symmetric slice is $m = \frac{r}{2}(1 - r'^2)$. The *magnetic self-energy* of that sphere is $\ell^2(1 - W^2)^2/(2r)$.

For the cap $r = \ell \sin(x/\ell)$ one finds $m = \frac{\ell}{2}\sin^3(x/\ell)$, and for the neck $r = \ell$ one finds $m = \ell/2$.

The slice is *conformally flat* when its Weyl curvature vanishes. In this sector the relevant invariant can be computed in double null coordinates, and it comes out as a difference of the two numbers above.

Proposition 16.2 (Weyl curvature = mass − self-energy). proved in Lean *In spherically symmetric Einstein–Yang–Mills theory the conformally invariant combination of the metric functions equals*

$$3\Big(\frac{2m}{r} - \frac{\ell^2(1-W^2)^2}{r^2}\Big).$$

Hence the slice is conformally flat exactly when $2mr = \ell^2(1-W^2)^2$: on every sphere the mass equals the magnetic self-energy.

In the repository: `CFT.DS.ds_weyl_mass`.

16.2 The action and the constraint

The Einstein–Yang–Mills action of Chapter 13, reduced to spherical symmetry, gives one equation that every slice must satisfy, the Hamiltonian constraint. At a moment of time symmetry it reads

$$-4rr'' + 2(1-r'^2) = 4\ell^2\Big(W'^2 + \frac{(1-W^2)^2}{2r^2}\Big), \qquad \ell^2 = \frac{4\pi G}{g^2}. \tag{16.2}$$

The left side is r^2 times the scalar curvature of the slice; the right side is r^2 times $16\pi G$ times the Yang–Mills energy density. You can check by hand that the cap $r = \ell\sin(x/\ell)$, $W = \cos(x/\ell)$ and the neck $r = \ell$, $W = 0$ both satisfy it: in both cases the two sides are $6\sin^2(x/\ell)$ and 2.

Multiplying (16.2) by $r'/4$ turns it into a statement about the mass:

$$m' = r'\,\ell^2\Big(W'^2 + \frac{(1-W^2)^2}{2r^2}\Big). \tag{16.3}$$

In the repository: `CFT.DS.ds2_mass_prime`. The mass grows wherever the sphere grows, at a rate fixed by the energy density.

16.3 A Bogomolny inequality

Now impose conformal flatness, $2mr = \ell^2(1-W^2)^2$, and differentiate it along the slice. Using (16.3) and dividing by $2\ell^2$ gives

$$rr'W'^2 + 2WW'(1-W^2) + \frac{(1-W^2)^2 r'}{r} = 0. \tag{16.4}$$

At first sight this is not illuminating. The trick, familiar from monopole physics, is to multiply by r' and look for squares.

Lemma 16.3 (completing the square). proved in Lean *For all real $r \neq 0$, r', W, W',*

$$r'\Big(rr'W'^2 + 2WW'(1-W^2) + \frac{(1-W^2)^2 r'}{r}\Big) = \frac{\big(rr'W' + W(1-W^2)\big)^2 + (1-W^2)^2\,(r'^2 - W^2)}{r}.$$

Lean: `ds2_bogomolny`

```
theorem ds2_bogomolny (r r' W W' : ℝ) (hr : r ≠ 0) :
    r' * (r * r' * W' ^ 2 + 2 * W * W' * (1 - W ^ 2)
          + (1 - W ^ 2) ^ 2 * r' / r)
      = ((r * r' * W' + W * (1 - W ^ 2)) ^ 2
          + (1 - W ^ 2) ^ 2 * (r' ^ 2 - W ^ 2)) / r := by
  field_simp
  ring
```

Theorem 16.4 (Bogomolny inequality for conformally flat slices)**.** proved in Lean *On a time-symmetric, conformally flat slice of spherically symmetric Einstein–Yang–Mills theory, at every point where $W^2 \neq 1$,*

$$r'^2 \leq W^2.$$

Equality holds if and only if $W = \pm r'$, and then the slice obeys the first-order equation $rr'W' = -W(1 - W^2)$.

Proof. By (16.4) the left side of the lemma vanishes, so the numerator on the right vanishes. It is a square plus $(1-W^2)^2(r'^2 - W^2)$ with $(1-W^2)^2 > 0$. If $r'^2 > W^2$ the numerator would be positive. At equality the second term is zero, so the square is zero. □

In the repository: `CFT.DS.ds2_num`, `ds2_W_ge`, `ds2_saturated`.

Remark. Earlier chapters imposed $W = r'$ by hand, motivated by the observation that the Yang–Mills potential can be identified with the spin connection of the slice (Chapter 15). Theorem 16.4 shows that this condition is not an extra assumption inside the conformally flat class: it is the case of equality of an inequality that every such slice satisfies. This is the same logic as the Bogomolny bound for monopoles, where the first-order equations describe the configurations that saturate an energy inequality.

16.4 The circle and its envelope

In the saturating case $W = r'$ we have $m = \frac{r}{2}(1 - W^2)$, and conformal flatness $2mr = \ell^2(1 - W^2)^2$ becomes $r^2(1 - W^2) = \ell^2(1 - W^2)^2$. Dividing by $1 - W^2$:

Theorem 16.5 (the shape equation)**.** proved in Lean *On a time-symmetric, conformally flat slice with $W = r'$ and $W^2 \neq 1$,*

$$r^2 + \ell^2 r'^2 = \ell^2. \tag{16.5}$$

In the repository: `CFT.DS.ds_circle_of_flat`.

Equation (16.5) says that the point $(r, \ell W) = (r, \ell r')$ moves on a circle of radius ℓ (Figure 16.2). The cap goes round half of it, from the north pole $(0, \ell)$ through the equator $(\ell, 0)$ to the south pole $(0, -\ell)$. The neck does not move at all: it sits at the single point $(\ell, 0)$.

Lean: `ds_cap_on_circle`

```
theorem ds_cap_on_circle (l x : ℝ) :
    (l * sin (x / l)) ^ 2 + l ^ 2 * cos (x / l) ^ 2 = l ^ 2 := by
  linear_combination l ^ 2 * sin_sq_add_cos_sq (x / l)
```

Equation (16.5) is of first order, and first-order equations can have solutions that are not members of the general family. Differentiating it gives $2r'(r + \ell^2 r'') = 0$, so at every point one of two things happens.

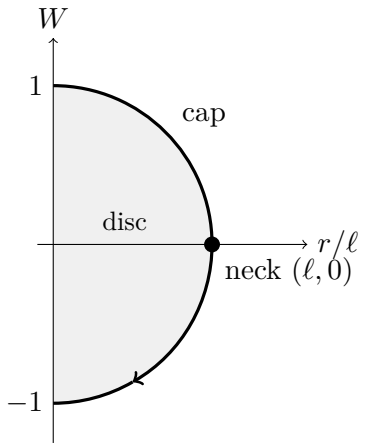


Figure 16.2: The (r, W) plane. Conformally flat slices with $W = r'$ live on the circle $r^2 + \ell^2 W^2 = \ell^2$. The cap runs along it; the neck is the point $(\ell, 0)$. The shaded disc is the region where conformally flat solutions have real null directions (Proposition 16.8).

Theorem 16.6 (two branches)**.** proved in Lean *Let r be twice differentiable and satisfy* (16.5)*. Then at every point*

$$r' = 0 \qquad \text{or} \qquad r'' = -\frac{r}{\ell^2}.$$

In the repository: `CFT.DS.ds_branch`.

The second branch is the harmonic equation; its solutions are the shifted caps $r = \ell \sin((x - c)/\ell)$, the regular solutions. The first branch, with $r^2 = \ell^2$, is the constant $r = \ell$: the neck. It is the *singular solution* of (16.5), the envelope of the family of caps (Figure 16.1). At $r = \ell$ the equation $r' = \pm\sqrt{\ell^2 - r^2}/\ell$ loses uniqueness, and this is the only place where a solution can leave the family of caps.

Corollary 16.7. *A C^1 curve that follows a cap and then the neck must switch at a point where the cap reaches $r = \ell$. There the first derivatives agree ($r' = 0$ on both sides) but the second derivatives are $-1/\ell$ and 0. The dumbbell is C^1 and never C^2.*

This is the geometric content of the "jump" that appeared in earlier chapters. It is not a defect put in by hand; it is the standard relation between a family of curves and its envelope: an envelope touches each curve, so slopes agree, but it does not bend with it.

Proposition 16.8 (the disc)**.** proved in Lean *For conformally flat solutions the slope dW/dr along a null direction solves a quadratic equation with coefficients $(1 - W^2)^2/r$, $2W(1 - W^2)$ and $(r^2 - \ell^2(1 - W^2)^2)/r$. Its discriminant is*

$$\frac{4(1 - W^2)^3}{r^2}\big(\ell^2(1 - W^2) - r^2\big),$$

so for $W^2 < 1$ real slopes exist exactly in the disc $r^2 + \ell^2 W^2 \le \ell^2$. The two roots coincide on the circle. The whole cap spacetime, $r = \ell \cos\eta \sin\chi$ and $W = \cos\chi$, lies in the disc, since $r^2 + \ell^2 W^2 = \ell^2 - \ell^2 \sin^2\eta \sin^2\chi$.

In the repository: `CFT.DS.ds_slope_disc`, `ds_slope_real`, `ds_cap_disc`.

16.5 Theorem A: the junction is forced

We can now put the pieces together. Once one decides to glue a cap to a neck, nothing about the gluing is left to choose.

Theorem 16.9 (uniqueness of the junction). proved in Lean *Consider the cap $r = \ell \sin(x/\ell)$ and the neck $r = \ell$ with $W = r'$.*

(i) The areal radius is continuous only at the equator $x = \pi\ell/2$ (`ju_location`).

(ii) There $r' = \cos(x/\ell) = 0$, so the first derivatives agree automatically (`ju_C1_auto`).

(iii) The second derivatives are $-1/\ell$ on the cap (`ju_cap_second`) and 0 on the neck, and at $r = \ell$, $r' = 0$ the Hamiltonian constraint (16.2) reduces to $r''(r'' + 1/\ell) = 0$, whose only roots are these two values (`rr_two_roots`).

(iv) The junction cannot be a timelike tube: along it the cap's equatorial radius $\sqrt{\ell^2 - \tau^2}$ differs from ℓ for $\tau \neq 0$ (`rj_tube_mismatch`).

(v) Along a null front the jump $[W_u]$ keeps its value at the corner, $1/(\sqrt{2}\,\ell)$ (`ju_jump_fixed`).

The combined statement is `CFT.JU.ju_theorem_A`.

Part (i) surprised the author. One would expect the place of the junction to be a free choice. It is not: the largest radius of the cap is exactly the radius of the neck, so they can touch only at the top of the cap, and there the slopes agree by themselves. "Glue continuously" already implies "glue in a C^1 way".

Where this stands

Known. The cap (the closed radiation universe driven by the homogeneous Yang–Mills field) and the neck (the magnetic Bertotti–Robinson spacetime) are classical solutions [1, 2, 3]. That the throat of an extremal black hole has a free length, and that it saturates an area–charge inequality, is known for Einstein–Maxwell theory [5, 6]. The relation between the Weyl curvature and the Misner–Sharp mass is known in general [4].

Claimed as new, in the weak sense of the guide for the specialist. The Bogomolny-type inequality $r'^2 \leq W^2$ for conformally flat slices (Theorem 16.4), the reading of the neck as the singular solution and envelope of the caps (Theorems 16.5 and 16.6), and the statement that the junction is forced (Theorem 16.9).

Assumptions to keep in mind. Spherical symmetry, a purely magnetic field, time symmetry and conformal flatness. Outside the class $W = r'$ the two roots of (iii) disappear, so these theorems do not exclude a smooth interpolation in general. Inside the conformally flat class a numerical search for a smooth static interpolation failed negative result (Chapter 19).

Open. Why the dumbbell is glued at all is not answered by these theorems; they say only how it must be glued if it is. Chapter 26 shows that a fixed total volume $2\pi^2\ell^3 + 4\pi\ell^2 L$ fixes the neck length L, and Chapter 23 shows that quantum mechanically L is conjugate to $-r\dot{r}$. open

Exercises

Each exercise is answered by the named Lean theorem. Read its statement in the repository and compare it with your own answer.

Exercise 16.1. Check by hand that the cap $r = \ell \sin(x/\ell)$, $W = \cos(x/\ell)$ lies on the circle (16.5), and that the neck $(r, W) = (\ell, 0)$ lies on it too. *Answer:* `ds_cap_on_circle`, `ds_neck_on_circle`.

Exercise 16.2. Multiply the Hamiltonian constraint (16.2) by $r'/4$ and show that it becomes (16.3). *Answer:* `ds2_mass_prime`.

Exercise 16.3. Expand both sides of the completing-the-square lemma and check that they agree. Where is $r \neq 0$ used? *Answer:* `ds2_bogomolny`.

Exercise 16.4. Theorem 16.5 assumes $W^2 \neq 1$. Find what goes wrong in the proof when $W^2 = 1$, and describe the slices with $W \equiv \pm 1$. *Answer:* the division in `ds_circle_of_flat`; for the slices, `fam_birkhoff_u` in Chapter 21.

Exercise 16.5. Show that $\ell \sin(x/\ell) = \ell$ with $0 < x < \pi\ell$ forces $x = \pi\ell/2$. *Answer:* `ju_location`.

Exercise 16.6. Compute the derivative of $\cos(x/\ell)$ at $x = \pi\ell/2$. *Answer:* `ju_cap_second`.

Exercise 16.7. Solve $2\ell^2 y^2 + 2\ell y = 0$ and explain why its two roots are the second derivatives of the neck and of the cap at the junction. *Answer:* `rr_two_roots`.

Exercise 16.8. Show that for $W^2 < 1$ the discriminant of Proposition 16.8 is non-negative if and only if $r^2 + \ell^2 W^2 \leq \ell^2$. *Answer:* `ds_slope_real`.

Chapter 17

Junctions

The question

When two different solutions of Einstein's equations meet along a surface, what must happen at the surface? In general a thin shell of matter has to sit there. For the dumbbell there are three kinds of surface, and the answer differs for each: a timelike junction needs both a gravitational shell and a colour current and is excluded, a junction inside a single slice needs a shell density, and a null front needs nothing at all. That is why the dumbbell evolves through null fronts.

17.1 Equations as distributions

A junction is a place where the metric is continuous but its derivatives jump. Einstein's equations then contain products like "derivative of a jump", which are delta functions. The clean way to handle them is to read the field equations as distributions: integrate against a smooth test function, move derivatives onto the test function by integrating by parts, and compare the coefficients of δ on both sides.

The repository does this from scratch. If r is continuous across x_0 and r' jumps by $[r']$, then in the weak sense r'' equals its regular part plus $[r']\,\delta_{x_0}$ (`weak_second_derivative`). The coefficient of δ is unique (`delta_coeff`, `rj_delta_coeff_one`), so one can read off what must sit on the surface.

17.2 Three kinds of surface

Theorem 17.1. proved in Lean *For spherically symmetric Einstein–Yang–Mills fields joined across a sphere:*

(i) Within a slice. *A jump in r' requires a shell of surface density $\sigma = -[r']/(4\pi G r_0)$; there is no shell exactly when r' is continuous.*

(ii) Timelike. *A timelike junction surface between a cap and a neck requires both a gravitational shell (a jump in extrinsic curvature) and a surface colour current. There is no shell-free timelike junction.*

(iii) Null. *A null front carries neither a shell nor a colour current, but W_u may jump; the jump is constant along the front.*

In the repository: `shell_density`, `no_shell_iff`, `timelike_no_go`, `dumbbell_no_shellfree_timelike`, `dumbbell_color_current`, `null_front_no_shell`, `jump_transport`, and the real-function versions `rj_tube_mismatch`, `rj_Wu_transport`.

Part (iii) is the reason the dumbbell has to be organised around light. The cap and the neck are glued at $x = \pi\ell/2$ on the initial slice. From there the influence of the junction travels outward along two null fronts, and on those fronts nothing singular sits: only the derivative of W across the front is discontinuous, a weak discontinuity of the kind that hyperbolic equations always transport along characteristics.

17.3 The general Israel condition

For non-null surfaces without any symmetry, the jump in extrinsic curvature is tied to the surface stress by Israel's condition, $S_{ab} = -\frac{1}{8\pi G}([K_{ab}] - h_{ab}[K])$. It can be inverted, $[K_{ab}] = -8\pi G\big(S_{ab} - h_{ab}\,\mathrm{tr}\,S/(n-1)\big)$, and there is no shell exactly when the extrinsic curvature is continuous (`israel_trace`, `israel_inverse`, `israel_no_shell`). For null shells the analogue is the Barrabès–Israel condition, derived in the repository from the distributional uu equation (`null_shell_from_weak`, `null_no_shell_iff`).

17.4 The size of the jump

At the corner where the null front leaves the junction sphere, the cap has $W_u = 1/(\sqrt{2}\ell)$ and the neck has $W_u = 0$. Since the jump is constant along the front, it stays $1/(\sqrt{2}\ell)$ all the way (`rj_capWu_corner`, `ju_jump_fixed`). The focusing equation then forces a jump in r_{uu} as well, $[r_{uu}] = -(2/r)[W_u^2]$ (`ruu_jump`, `rj_ruu_corner`).

A correction recorded honestly

An earlier version of these theorems passed from abstract derivations to real functions through a structure called `RealLine`. It turned out that its hypotheses could only be satisfied by constant functions, so the final theorems were vacuous. Version 0.6.0 proves them again directly for real functions with ordinary derivatives (`CFT.RJ`). The episode is kept in the record because it is the kind of error a proof assistant makes visible.

Where this stands

The Israel and Barrabès–Israel conditions are classical. The analysis of the three junction types for the cap–neck pair, the impossibility of a shell-free timelike junction, and the transport of the jump $1/(\sqrt{2}\ell)$ along the null fronts are results of the programme, checked here from the distributional equations.

Exercises

Exercise 17.1. Show that r'' contains $[r']\delta$ in the weak sense. *Answer:* `weak_second_derivative`.

Exercise 17.2. Invert Israel's condition. *Answer:* `israel_inverse`.

Exercise 17.3. Show that $[W_u]$ is constant along a null front. *Answer:* `rj_Wu_transport`.

Chapter 18

Time evolution

The question

Released from its initial slice, what does the dumbbell do? The caps collapse to a curvature singularity at proper time ℓ. The neck does not collapse at all. Between them a wedge bounded by the two null fronts evolves by its own equations, and numerically it falls into a singularity of its own after about $1.27\,\ell$.

18.1 The constraints are preserved

Before evolving anything one must know that the constraints of Chapter 11, once satisfied, stay satisfied. In synchronous gauge with the full spherically symmetric Einstein–Yang–Mills system, the time derivatives of the Hamiltonian and momentum constraints are linear combinations of the constraints themselves (`hamiltonian_propagation`, `momentum_propagation`). In double null coordinates the same holds for the two focusing constraints, and in Chapter 25 we will see that it continues to hold when fermions are added.

18.2 The cap collapses

The cap is the closed radiation universe $a = \ell \cos\eta$. Its curvature can be measured by the Kretschmann invariant $K = R_{abcd}R^{abcd}$, computed in the repository from the Riemann tensor of the metric.

Theorem 18.1. proved in Lean *On the cap, $K = 24\ell^4/a^8$. As $\eta \to \pi/2$, $a \to 0$ and $K \to \infty$. Since K is independent of position, this happens everywhere on the cap at once, including its centre, at central proper time $\tau = \ell$.*

In the repository: `kret_cap`, `kret_blowup`, `center_limit`.

Light emitted from the junction at $\chi = \pi/2$ travels inward along $\chi = \pi/2 - \eta$ and reaches the centre exactly at the crunch (`front_hits_center`). The centre never learns that a neck was attached.

18.3 The neck does not

On the neck the natural clock is the proper time t of observers who start at rest on the initial slice. In these Gaussian coordinates the neck metric is $-dt^2 + \cos^2(t/\ell)\,dz^2 + \ell^2 d\Omega^2$ (`neck_gaussian`). At $t = \pi\ell/2$ the factor $\cos^2(t/\ell)$ vanishes: neighbouring geodesics meet. This is a focal point of the time slicing, not a curvature singularity, since the neck's curvature is constant.

For a while it was tempting to identify the cap's conformal time with the neck's time and see the crunch and the focusing as one event. That is wrong: in the common Gaussian clock the cap crunches at ℓ and the neck focuses at $\pi\ell/2$, two different events (`two_events`). negative result

18.4 The wedge

Between the two null fronts lies a region that is neither cap nor neck. Its evolution is a characteristic initial value problem: data on the two fronts determine the solution inside. Local existence and uniqueness for such problems are classical theorems external theorem. Numerically, with the equations in double null form, the following was found numerical:

- the constraints are preserved with second-order convergence;
- the jump $[W_u]$ along the fronts stays at $1/\sqrt{2}$ in units of ℓ;
- the areal radius reaches zero on a spacelike curve; the region between the fronts survives for about $1.27\,\ell$ of proper time;
- the wedge is a trapped region: both null expansions are negative there (`oc_trapped`).

18.5 The neck's throat under a source

If a transverse wave passes through the throat, the throat itself can collapse. With a source of strength $(rW)^2 \geq \kappa > \ell^2/2$, starting from rest at radius r_1, the throat radius decreases monotonically and reaches zero before half the natural period (`acc_le`, `rp_nonpos`, `z9_collapse`, `z9_before_half_period`). The proof derives the monotonicity from the equation rather than assuming it.

Where this stands

The cap's collapse is that of Tolman's universe, and the constraint propagation is standard. The common-clock correction, the trapped wedge and the throat collapse theorem are part of the programme. The wedge's lifetime is a numerical result. Whether the singularity of the wedge is the endpoint of the dumbbell or the place where quantum gravity takes over is open. open

Exercises

Exercise 18.1. Show that $K = 24\ell^4/a^8$ on the cap. *Answer:* `kret_cap`.

Exercise 18.2. Show that the inward null front from $\chi = \pi/2$ reaches $\chi = 0$ only at $\eta = \pi/2$. *Answer:* `front_hits_center`.

Exercise 18.3. Show that the cap crunch and the neck focus are distinct events. *Answer:* `two_events`.

Chapter 19

The conformal frame and Ω

The question

The cap and the neck look very different: one is a shrinking sphere, the other an endless tube. Is there a frame in which they look alike? Yes. Both are conformally the Einstein static universe $\mathbb{R} \times S^3$, multiplied by a single function Ω, and in that frame Einstein's equations collapse to the linear equation $\hat{\Box}\Omega = \Omega$. This chapter also contains the linear rigidity theorem, which the author believes to be new.

19.1 Why a conformal frame

Yang–Mills theory in four dimensions does not care about the overall scale of the metric. If $g = \Omega^2 \hat{g}$, a Yang–Mills field on g is also one on $\hat{g}$, and its stress is traceless. Choose $\hat{g} = -d\eta^2 + du^2 + \sin^2 u \, d\Omega^2$, the static universe whose space is the unit three-sphere: the quantum sphere of Chapter 6, now as space. The trace of Einstein's equations says $R[g] = 0$. The conformal transformation formula for the scalar curvature (quoted external theorem) then turns this into

$$\hat{\Box}\,\Omega = \Omega,$$

a linear wave equation for the conformal factor. That is a remarkable simplification for a non-linear theory.

19.2 Cap and neck as conformal factors

Theorem 19.1. proved in Lean *The cap is $\Omega = \ell \cos\eta$ (depending only on time) and the neck is $\Omega = \ell / \sin u$ (depending only on position). Both satisfy $\hat{\Box}\Omega = \Omega$. On both, the $\eta\eta$ component of Einstein's equations reduces to $\ell^2 = 4\pi G / g^2$.*

In the repository: `cap_box`, `neck_box`, `eta_cap`, `eta_neck`.

The neck's coordinate $x = \ell \log\tan(u/2)$ turns $(\ell^2/\sin^2 u)(du^2 + \sin^2 u \, d\Omega^2)$ into $dx^2 + \ell^2 d\Omega^2$ (`neck_coordinate`): the two infinitely distant ends of the neck are the two poles of S^3. At the junction $u = \pi/2$ both factors equal ℓ with zero slope, and the second derivatives are 0 and ℓ (`junction_seam`). The C^2 seam of Chapter 16 is a seam in Ω''.

The areal radius is $r = \Omega \sin\chi$, and it satisfies the flat two-dimensional wave equation (`oq_radius_wave`). So $r = F(\eta + \chi) - F(\eta - \chi)$ for some function F, and the antipodal map of S^3 sends Ω to $-\Omega$ (`oq_antipodal`).

19.3 Linear rigidity

Here is the theorem that the author submitted for publication.

Theorem 19.2 (linear rigidity). proved in Lean *Every linear perturbation of the Einstein–Yang–Mills Tolman universe that keeps spherical symmetry, the purely magnetic ansatz and conformal flatness is generated by a conformal Killing vector of the Einstein static universe. In particular it is pure gauge.*

In the repository: the three generating modes `rr_mode0`, `rr_mode1`, `rr_mode2`, the integrability condition `rr_eta_integrability`, the reduced equation `rr_chi_ode` and the exclusion of the remaining mode `rr_cos_mode_excluded`.

The proof is local and uses neither mode expansions nor boundary conditions. The perturbations of Ω and W must satisfy the linearised constraints; integrating them in η and χ leaves a three-parameter family, and the three parameters are exactly the three conformal Killing vectors ∂_η, ξ_1, ξ_2 of the static universe that preserve spherical symmetry. They close into the algebra $\mathfrak{sl}(2,\mathbb{R})$ (`oq_sl2_bracket`), and the neck is the unique configuration invariant under all three (`oq_neck_invariant`, `oq_neck_unique`).

This does not contradict the known instabilities of the Einstein static universe or of Yang–Mills fields on a fixed background, because those perturbations leave the conformally flat class. The rigidity theorem says that within this class there is nothing to perturb.

19.4 The self-dual connection

Chapter 4 split Lorentz transformations into rotation and boost. General relativity has a matching split: the self-dual connection $A = \Gamma + iK$ combines the spin connection Γ of space with its extrinsic curvature K. On the cap its coefficient is $c = 1 + i\dot{a}$, and the curvature coefficient $c^2 - 2c = -|c|^2$ is real. The Friedmann equation becomes $|c|^2 = (8\pi G/3)\rho a^2$, and the phase of c is the conformal time (`curvature_real`, `friedmann_selfdual`, `phase_lemma`). On both exact solutions the Yang–Mills field equals the real part of A (`cap_real_part`, `real_part`).

19.5 A tempting quantisation, and why it was withdrawn

The cap's action in the conformal frame is $S = (3\pi/4G)\int(\Omega^2 - \dot{\Omega}^2)\,d\eta$, a harmonic oscillator of period 2π (`periodic`, `energy_balance`). Quantising it naively gives $\ell^2/\ell_P^2 = 4(n+\frac{1}{2})/(3\pi)$ (`quantized`), and with it a candidate value of the coupling. This was later withdrawn: in the full conformally flat magnetic sector the constraints leave no local degree of freedom, the quantum theory of that sector has a single state, and the coupling is not quantised (Chapter 23). negative result

Where this stands

The conformal transformation formula is quoted. The rigidity theorem is claimed as new (claim 1 of the guide for the specialist); a manuscript containing it was submitted to a journal and returned without review. The self-dual description is checked on the exact solutions only; in general evolution the statement "Yang–Mills equals the real part of A" is a postulate, not a theorem.

Exercises

Exercise 19.1. Show that $\ell\cos\eta$ and $\ell/\sin u$ satisfy $\hat{\Box}\Omega = \Omega$. *Answer:* `cap_box`, `neck_box`.

Exercise 19.2. Compute the three brackets of the conformal Killing vectors. *Answer:* `oq_sl2_bracket`.

Exercise 19.3. Show that $(1-x^2)U''-5xU'+n(n+4)U=0$ implies the Yang–Mills test-field frequencies $\omega^2=(n+2)^2-3$. *Answer:* `oq_spectrum`.

Chapter 20

Stability

The question

Is the dumbbell stable? Not as it stands. The neck carries a mode of the gauge field below the Breitenlohner–Freedman bound, and a long enough neck is certainly unstable. The cap sits on top of a potential barrier. An adjoint Higgs field heavy enough can cure the neck, and this gives the stability bound that will meet the weak gravity conjecture in Chapter 26.

20.1 The Breitenlohner–Freedman bound

In anti-de Sitter space a field can have a negative mass squared and still be stable, as long as $m^2\ell^2 \geq -1/4$ in two dimensions. Below that bound perturbations grow. For a mode e^{-ks} the characteristic equation $k^2 - k - m^2\ell^2 = 0$ has real roots exactly when $m^2\ell^2 \geq -1/4$ (`char_root`, `real_roots_iff`, `bf_ads2`).

On the neck, the gauge-field mode W has $m^2\ell^2 = -1$. It violates the bound, and inside the throat its profile oscillates as $e^{-s/2}\cos(\sqrt{3}\,s/2)$ (`throat_mode`, `bf_violation`).

20.2 How long a neck is unstable

Pure AdS_2 has no length scale, so the rate of growth must come from where the neck is cut off, at the junction (`no_intrinsic_rate`). A Liouville transformation turns the quadratic form of the mode into $\int (h'^2 - \frac{3}{4}h^2)\,dy$ over an interval of length Y, which is negative for a suitable trial function when $Y > 2\pi/\sqrt{3}$ (`Q_value`, `Q_neg_iff`).

Theorem 20.1. proved in Lean *If the half-length L of the neck exceeds $\ell\,(2\pi/\sqrt{3}+\log 2) \approx 4.32\,\ell$, the neck has a growing mode.*

In the repository: `Y_lower`, `long_neck_unstable`, and the sharper version with the actual potential `long_neck_unstable_actual`.

This is a sufficient condition. Whether short necks are stable is open. open

20.3 The cap is a sphaleron

On the cap the gauge field is homogeneous, $W = f(\eta)\cos\chi$ in a suitable sense, and the reduced dynamics is a double well $(1-f^2)^2$ with the cap sitting exactly on top of the barrier, at $f = 0$. Small deviations grow at rate $\sqrt{2}$ (`ol_sphaleron_rate`). The geometry does not notice: the conformal factor obeys $\Omega'' = -\Omega$ whatever f does, because the stress stays that

of radiation (`ol_geometry_rigid`). The field is unstable; the spacetime is rigid. This is the dynamical counterpart of the linear rigidity theorem.

20.4 Curing the neck with a Higgs field

Add an adjoint Higgs field with vacuum value v. The neck remains an exact solution with the same radius (`neck_vacuum`, `force_W_zero`), and the W mode acquires a mass from the Higgs mechanism, so that $m^2\ell^2 = -1 + m_W^2\ell^2$ (`w_mass_neck`).

Theorem 20.2. proved in Lean *If $m_W^2\ell^2 \geq 3/4$, equivalently $v \geq \sqrt{3/(16\pi)}\, M_{\mathrm{Pl}} \approx 0.244\, M_{\mathrm{Pl}}$, the W mode has no growing mode on a neck of any length.*

In the repository: `bf_condition`, `planck_condition`, `hardy_neck`, `no_growing_mode`.

The proof uses a Hardy-type inequality on the neck rather than the BF bound of the infinite throat, so it holds for finite necks too.

20.5 The smoothness map

What smooth static configurations exist at all? With a cosmological constant there is a family of cylinders of radius R with $\Lambda = 1/R^2 - 1/R^4 \leq 1/(4\ell^2)$, the maximum at $R = \sqrt{2}\,\ell$ (`cylinder_family`, `cylinder_max`). At a static throat $r'' = -2W'^2/r \leq 0$, so a static shape cannot have a waist (`no_static_throat`). The Einstein static universe with a Yang–Mills field is an exact solution with radius $\sqrt{2}\,\ell$ and $\Lambda = 3/(4\ell^2)$ (`esu_ym`). Numerically, the known smooth static sequences approach a smooth core joined to an infinitely long throat of radius ℓ numerical: smooth solutions exist, but they are not dumbbells of finite length.

Where this stands

The BF bound and the instabilities of Yang–Mills fields in AdS_2 are known. The length threshold, the sphaleron reading of the cap with its rigid geometry, and the Higgs stability bound with its Planck-scale vacuum value are results of the programme. The bound $m_W^2\ell^2 \geq 3/4$ reappears in Chapter 26 as one side of a window whose other side is the weak gravity conjecture.

Exercises

Exercise 20.1. Show that $k^2 - k - m^2\ell^2 = 0$ has real roots iff $m^2\ell^2 \geq -1/4$. *Answer:* `real_roots_iff`.

Exercise 20.2. Check that $e^{-s/2}\cos(\sqrt{3}s/2)$ solves $W'' + W' + W = 0$. *Answer:* `throat_mode`.

Exercise 20.3. Show that $1/R^2 - 1/R^4 \leq 1/4$ with equality at $R = \sqrt{2}$. *Answer:* `cylinder_max`.

Interlude B

Contact with string theory

The question

Does the dumbbell know anything about string theory? A few things fit together surprisingly well: the three-sphere of the cap is a consistent string background, the gravitational corrections of heterotic string theory cancel on the dumbbell, and the neck is exactly an attractor geometry. One thing does not fit: the natural string level comes out fractional. This interlude reports both.

This interlude can be skipped. It is placed after Part IV because it uses the neck and its stability.

B.1 The three-sphere as a string background

A string can propagate consistently on $S^3 = SU(2)$ if the curvature of space is balanced by a three-form flux H. The required condition at one loop is $\mathrm{Ric}_{ab} = \frac{1}{4} H_{acd} H_b{}^{cd}$. On $SU(2)$ the parallelising torsion of the left-invariant frame supplies exactly such an H, and the identity holds term by term, using only the quaternion commutators $[i, j] = 2k$ and their cyclic images (`quat_bracket`, `eps_contract`, `s3_ricci`, `parallelizing_torsion`). The total flux through a sphere of radius R is $4\pi^2 R^2$ (`flux`).

For the Wess–Zumino term to be well defined, the flux must be quantised in units of α': $R^2 = k\alpha'$ with an integer level k (`level`, quoted external theorem). If one sets $R = \ell$ and uses the heterotic relation between G, g and α', one finds $k = c$ with $c = 1/8$ or $1/4$ depending on conventions. Neither is an integer (`wzw_level`, `wzw_obstruction`). negative result

B.2 The standard embedding

Heterotic string theory corrects Einstein's equations at order α' by terms involving $\mathrm{tr}\, R^2 - \mathrm{tr}\, F^2$. If the gauge connection is set equal to the spin connection, the "standard embedding", these terms cancel. On the Euclidean dumbbell, where the spin connection lives in the compact algebra $\mathfrak{so}(4)$, the embedding is well defined and all α' terms in the equations of motion vanish (`gs_rhs_vanishes`, `all_alpha_terms_vanish`). This is the stringy echo of $W = r'$: the gauge field is the spin connection.

B.3 The neck is an attractor

Sen's entropy function method finds the near-horizon geometry of an extremal black hole by extremising a function of the sizes v_1 of AdS_2 and v_2 of S^2. For Einstein–Yang–Mills with

magnetic charge p it is

$$\mathcal{E}(v_1, v_2) = \frac{\pi}{G}(v_2 - v_1) + \frac{4\pi^2 p^2}{g^2}\frac{v_1}{v_2},$$

derived from the action (`entropy_from_action`). Its only critical point is $v_1 = v_2 = p^2\ell^2$, the critical value equals the Bekenstein–Hawking entropy, and for $p = 1$ the sphere has radius exactly ℓ (`z41_critical`, `z41_entropy`, `z41_neck`). The neck of the dumbbell is the attractor.

B.4 The dumbbell is not supersymmetric

If a Killing spinor existed, its bilinear would be a timelike Killing vector, and at the throat it would force $\partial_t r = 0$. But the throat collapses under a source (Chapter 18). So the collapsing dumbbell admits no Killing spinor (`killing_time_pos`, `no_killing_spinor`, `no_killing_spinor_cap`). Consistently, supersymmetric fluctuation operators have the form $Q^\dagger Q + QQ^\dagger$ and no negative modes (`susy_posSemidef`, `no_bf_violating_mode`), while the neck has one.

Where this stands

WZW models, the standard embedding and the attractor mechanism are standard string theory. Their application to the dumbbell is checked here. The fractional level is a genuine obstruction to embedding the dumbbell with $R = \ell$ into an $SU(2)$ WZW model and is recorded as a negative result.

Exercises

Exercise B.1. Verify $\sum_{cd} \varepsilon_{acd}\varepsilon_{bcd} = 2\delta_{ab}$. *Answer:* `eps_contract`.

Exercise B.2. Find the critical point of $\mathcal{E}$. *Answer:* `z41_critical`.

Exercise B.3. Show that a level k with $0 < k < 1$ is not a positive integer. *Answer:* `wzw_obstruction`.

Part V

Heat and quanta

Part VI

Shape, swampland and open problems

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